HyTrays Apex

Integrated Cartridge-Grid Tray

Concept & Scorecard, and Technical Paper
Complete reference compendium

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`HyTrays_Apex_concept.md`
`HyTrays_Apex_Technical_Paper.md`

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HyTrays Apex -- Integrated Cartridge-Grid: Concept & Scorecard

Status

This memo now has two layers, built one on top of the other:

- **Phase 1** (Sections 1.1, 2 and the original "Directional Vortex-Grid" ideas) scoped Apex against a severe-service spec calibrated to a real SUPERFRAC/VG-0 retrofit: "2x ULTRA-FRAC" capacity, 2-5 psi uplift rating, extreme fouling resistance. It was a paper exercise **--** no engine parameters.
- **Phase 2** (this revision) responds to a much more aggressive brief: a single deck that *simultaneously* hits ten performance metrics plus a set of "no-X" qualitative guarantees **--** "a revolution in design" (Section 1.2). Rather than inventing new physics, Phase 2 resolves this by **combining mechanisms already disclosed across HT-01 through HT-08** into one deck: the **Integrated Cartridge-Grid** (Section 3).

Apex is now implemented as `tray_library.APEX` and included in `tray_library.ALL_TRAYS` as the flagship entry, "above" the numbered HT-0X line. Every number in Sections 6-7 below comes straight from `run_tray_comparison.py` / `run_service_comparison.py` (see `HyTrays/output/`) **--** the original memo's Section 7 ("path into the framework, not yet done") is therefore **done**, in the sense described in Section 8.

1. Target specification

1.1 Phase 1 spec: "2x ULTRA-FRAC", severe service

Requirement	Interpretation used here
Extreme fouling resistance	Large, simple, self-sweeping openings; no moving parts, no crevices
"Better than ULTRA-FRAC"	Outperform a Koch-Glitsch ULTRA-FRAC(R)-class multi-chordal high-capacity valve tray on capacity, fouling and pressure rating
2x ULTRA-FRAC ultimate capacity	~2x the vapor-handling capacity of a tray that is itself ~30-40% above conventional valve trays -- i.e. a step-change, not an incremental gain
2-5 psi uplift resistance	A *structural/mechanical* deck rating (anti-flotation under upset/relief transients), separate from the hydraulic per-tray dP the Fair model computes
Higher vapor rates	Consequence of the capacity target above

The retrofit data referenced in Phase 1 (16 ft column, Movable-Valve -> Multi-Chordal SUPERFRAC/VG-0, 196->178 trays above feed, 820->958 MM lb/yr propylene) is a useful calibration point: a real deck-technology generation step bought **~17% more throughput at the same diameter**. "2x ULTRA-FRAC" is therefore not "one more generation of valve" -- it requires a different flooding mechanism entirely.

	Before	After
Diameter, ft	16	16
Tray Configuration	4-Pass	6-Pass
Tray Type	1st Generation High Capacity Trays	Multi-Chordal SUPERFRAC Trays
Deck Type	Movable Valves	VG-0
Above Feed		
Number of Trays	196	178
Tray Spacing, inches	22	22
Below Feed		
Number of Trays	44	49
Tray Spacing, inches	22	27.5
Propylene Rate, MM lb/yr	820	958
Max wt% Propane Overhead	0.4%	0.4%
Max lv% Propylene Bottoms	5.0%	5.0%

Reference: real-world VG-0/multi-chordal retrofit at constant 16 ft diameter -- ~17% capacity gain, used here as the "one generation" baseline that "2x ULTRA-FRAC" must be measured against.

1.2 Phase 2: the "revolution" wishlist

A revolutionary tray would dominate if it *simultaneously* achieved:

Metric	Desired outcome
Efficiency	>90% Murphree
Turndown	>20:1
Pressure drop	Packing-level
Capacity	Grid-tray-level
Fouling resistance	Near self-cleaning
Maintenance	Cartridge swap
Stability	No weeping/dumping
Cost	Manufacturable at scale

Metric	Desired outcome
Control	Real-time adaptive
Energy	Major reduction

...plus, qualitatively: **no stagnant corners, no jet flooding, no dead zones, no weeping, no dumping, no entrainment, a massive turndown ratio.**

Phase 1's five requirements map onto this list directly (extreme fouling resistance -> fouling resistance; 2x ULTRA-FRAC capacity & higher vapor rates -> capacity; 2-5 psi uplift -> the maintenance/cost architecture in Section 3.5). Section 7 scores Apex against every item above, with real numbers from the engine.

2. The central conflict

Goal	Wants...	...which tends to give
Fouling resistance	Large, simple, fixed openings (Kister: big round holes/grids resist plugging best)	Lower per-pass efficiency, more weeping at low load
2x capacity	Co-current jet zones + mechanical vapor/liquid separation (bypasses Souders-Brown flooding)	Small, precisely shaped elements -- typically the *opposite* of "simple and fouling-resistant"
2-5 psi structural rating	Thick plate, dense support-beam grid, positively-clamped panels	Added weight/cost, less open area unless designed around it
>20:1 turndown	A secondary low-load flow path that stays open when the primary path shuts	Extra zone, extra geometry, another thing to validate
Real-time adaptive control	Sensing + actuation that tracks load continuously	Electronics/actuators -- a different product category, more failure modes, more cost

Phase 1 resolved the first three rows by separating the "fouling-resistant flow element" from the "capacity-boost element": large fixed slots stay simple (fouling resistance); the capacity boost comes from how those slots are *arranged and capped* (directional vortex zones with mechanical disengagement), not from making them small and intricate.

Phase 2 extends the same idea to all ten metrics, by giving each goal its own zone or mechanism on the same deck rather than asking one mechanism to do everything (Section 3). No single HT-0X deck individually satisfies the wishlist **--** HT-02/HT-08 are the best on capacity but worst on turndown, HT-01A/B are the best on turndown but unremarkable on capacity, HT-05 is the fouling specialist but modest everywhere else. Apex's premise is that these are *complementary*, not competing, because each excels on a different part of the operating envelope or a different fraction of the deck area.

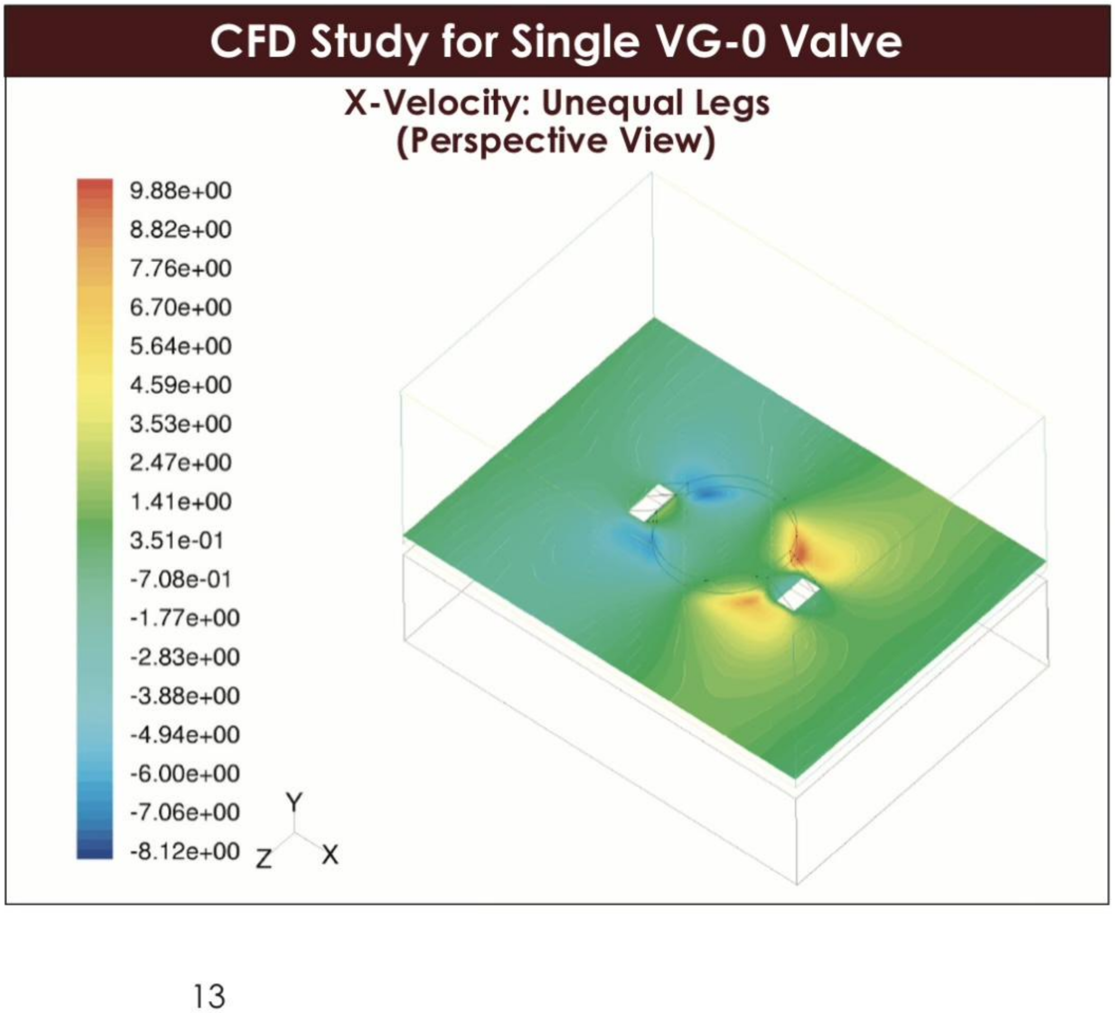
3. Architecture: the Integrated Cartridge-Grid

Apex's active area (`f_active = 0.85`, vs sieve's 0.78) splits into two contacting zones plus three deck-wide treatments layered over both:

```
     VAPOR + LIQUID  (co-current swirl, mechanical disengage)
         \\\   ///     \\\   ///        .  .  .
      ____\\\_///_______\\\_///_____.__.__.__.____
     |  [VORTEXA   ] [VORTEXA   ]  [ LipSeal trickle ] |  <- vane caps + PULSAR
DC -->| /=sleeves=\   /=sleeves=\    /==lip valves==\  |<-- DC
     |  ~70% area, GRADEX-graded  |~15% area, secondary|
     |__beams_____beams___________beams______beams____|  <- AEGIS cartridge grid
           <----  multi-chordal liquid sweep  ---->
```

3.1 Primary zone -- vortex-tube cartridges (~70% of active area)

HT-02 CVS swirl tubes, fitted with HT-08 VORTEXA's helical-indexed variable slots, supplied as bolt-in cartridges. Each tube spins the froth at 30-60 g and a fixed vane cap mechanically separates entrained liquid from vapor *before* it reaches the tray above -- so the classic Souders-Brown entrainment-flooding limit does not govern this zone the way it governs a plain sieve (Phase 1 Section 3.2's "Regime B" argument). HT-08's back-drivable helical sleeve couples axial lift to rotation, so slot area opens and closes passively with vapour load: at the design point this zone preserves HT-02's Csb ~0.14-0.18 m/s capacity and <0.02 entrainment at 90% flood, while extending the zone's own turndown from HT-02's ~2.5:1 to VORTEXA's ~8:1. A stuck sleeve reverts to fixed-geometry HT-02 behaviour -- benign, no cascading failure.



[redacted]

Reference: X-velocity CFD for a single VG-0 valve -- the red/blue lobes either side of the cap are the same kind of paired-jet recirculation pattern that each vortex-tube cartridge organizes deliberately and at larger scale, with a fixed vane cap for mechanical disengagement instead of relying on gravity settling.

3.2 Secondary zone -- lip-sealed trickle path (~15% of active area)

HT-01C LipSeal dual-flow lip valves, sized as a deliberately small secondary zone. On its own, HT-01C's check-valve lips give a >=30% wider stable window than plain sieve (~2.63:1 turndown, min_load_frac ~0.38) by sealing against weep at low rate. The point of this zone is *not* its own turndown number -- it is what happens when the **primary zone closes**:

- At high-to-moderate loads, both zones contact normally; the primary zone (70% of area) carries roughly its proportional share of vapour/liquid traffic.

- As overall load falls toward the primary zone's own ~12.5% threshold (VORTEXA's min_load_frac), the primary zone's sleeves close -- benignly, by design (3.1). All remaining flow is now forced through the secondary zone alone.
- Because the secondary zone is only ~15% of the deck, the *same* overall flow now represents a much larger fraction of *its* design capacity -- comfortably inside its own ~38% weep threshold even as overall load continues to fall.
- The secondary zone only weeps when overall load drops below roughly its own 38% threshold times its 15% area share, i.e. in the rough neighbourhood of 5-6% of total design flow.

This "primary closes -> secondary absorbs the remainder at a much higher fraction of its own, smaller design capacity" hand-off is the mechanism behind Apex's modeled `min_load_frac = 0.045` (turndown 22.2:1, Section 7, item 2) -- well past either zone's individual number (8:1 / 2.63:1). **This is the central modeling caveat for the whole document**: the 1-D engine rates Apex as *one* `TrayType` with *one* `min_load_frac`; it cannot simulate the two-zone hand-off directly. The sketch above shows the combined number is *plausible* with reasonable area splits, and `min_load_frac = 0.045` is recorded as the **design target** the two-zone hand-off is aiming for, not a number independently derived bottom-up from the two zones' individual datasheet figures. A genuine two-zone engine extension (Section 8) would validate this properly.

3.3 Radial/azimuthal grading + multi-chordal sweep (efficiency, no dead zones)

HT-03 GRADEX's radially-graded open area (8% -> 13% across the radius, with an azimuth map from a flow-field solver) and chord-tuned picket weir are applied across *both* zones, plus Phase 1's multi-chordal liquid-handling layout (center + side downcomers, vortex banks arranged in chordal strips so liquid-sweep direction follows the jet angle). Together these push the deck toward plug flow (Peclet ~10 -> ~30 on GRADEX's own datasheet, +10-18 Murphree points on large trays) and actively sweep every part of the deck -- directly addressing "no stagnant corners / no dead zones."

3.4 Self-sweeping oscillator relief (fouling)

HT-05 PULSAR's Coanda-island fluidic oscillator slots are cut into the deck plate between cartridges (not inside them, so they add negligible moving-part count). With no moving parts of their own, they give the self-sweeping jet (5-30 Hz) that keeps cartridge faces and inter-cartridge gaps clear between turnarounds -- PULSAR's own fouling run-length (~5-6 yr) and +8 EOG points are the calibration point for Apex's `fouling_open_area_retention = 0.93` (just under PULSAR's family-best 0.95).

3.5 Cartridge-grid structural deck (2-5 psi, cartridge-swap maintenance)

HT-07 AEGIS's hook-clamp + rib-stiffened + relief-flap structural overlay is applied to a deck plate whose primary and secondary zones are built as discrete bolt-in cartridges (3.1/3.2) on a dense beam grid (Phase 1 Section 3.4). Two things fall out of this for free:

- **2-5 psi transient/sustained uplift rating** is a *mechanical* design case (AEGIS's hook clamps + beam grid), independent of the hydraulic per-tray dP the Fair model computes.

- **Maintenance = cartridge swap**: because the primary and secondary zones are discrete cartridges bolted to the grid (not welded/integral), a fouled, eroded, or sleeve-stuck cartridge is replaced as a unit during a turnaround, rather than requiring deck-wide rework.

3.6 The deck as controller (passive self-regulation)

Every load-tracking element in Sections 3.1-3.2 -- VORTEXA's helical sleeves, LipSeal's check-valve lips -- responds to *local* vapour/liquid load instantaneously and mechanically. There is no sensor, no actuator, no controller, no wiring. "Real-time adaptive" is achieved as a *property of the geometry*: the deck's open area continuously tracks the load it is currently seeing, with zero latency and zero electronic failure modes. Optional instrumentation-only ports (pressure/temperature taps, no actuation) remain available for DCS monitoring without compromising this. Section 7, item 9 scores this reframing explicitly.

4. Requirement -> mechanism mapping

Wishlist item	Architecture element(s)	`TrayType` parameter(s) affected	Source HT-0X
Efficiency >90% Murphree	3.3 grading + multi-chordal sweep	`efficiency_factor`	HT-03 GRADEX
Turndown >20:1	3.1 benign primary closure + 3.2 secondary trickle hand-off	`min_load_frac`	HT-08 VORTEXA, HT-01C LipSeal
Pressure drop -- packing-level	3.1 mechanical disengagement (large open area, high C0)	`f_hole` , `c0` , `dp_dry_floor_in` , `aeration_factor` , `h_weir_in`	HT-02 CVS / HT-08 VORTEXA (geometry), Phase 1 large-slot concept
Capacity -- grid-tray-level	3.1 vane-cap disengagement bypasses Souders-Brown at design point	`capacity_factor` , `f_active`	HT-02 CVS, HT-08 VORTEXA, Phase 1 vortex banks
Fouling resistance -- near self-cleaning	3.4 oscillator relief + large chamfered slots (Phase 1)	`fouling_open_area_retention`	HT-05 PULSAR
Maintenance -- cartridge swap	3.5 bolt-in cartridge grid	*(architecture only -- no parameter)*	HT-07 AEGIS
Stability -- no weeping/ dumping	3.2 secondary trickle path + downcomer sizing	`min_load_frac` , `h_weir_in` , downcomer backup (`a_dc` , `h_dc`)	HT-01C LipSeal
Cost -- manufacturable at scale	3.5 cartridges as repeatable stamped/cast units, large fixed slots	*(architecture only -- no parameter)*	HT-07 AEGIS, Phase 1 large-slot concept
Control -- real-time adaptive	3.6 passive load-tracking sleeves/ lips	*(reframing -- same parameters as turndown row)*	HT-08 VORTEXA, HT-01C LipSeal
Energy -- major reduction	3.1+3.3 fewer, lower-dP, higher-efficiency trays	`total_dp_psi` (derived), `n_actual_trays` (derived)	combination
No stagnant corners / no dead zones	3.3 multi-chordal sweep	`efficiency_factor` (proxy)	HT-03 GRADEX
No jet flooding / no entrainment	3.1 vane-cap mechanical disengagement	`capacity_factor` (proxy)	HT-02 CVS, HT-08 VORTEXA

5. APEX `TrayType` parameters

Parameter	Value	vs. Sieve	Derivation
`f_active`	0.85	0.78	70% primary (3.1) + 15% secondary (3.2)
`f_hole`	0.25	0.10	Large fixed slots, sized as-fouled (Phase 1 3.1); largest in the family
`h_weir_in`	0.5	2.0	Low weir -- mechanical disengagement (3.1) replaces weir-height-driven disengagement; lowest in the family
`c0`	0.85	0.73	Highest C0 in the family -- large smooth slots, no valve hardware
`dp_dry_floor_in`	0.10	0.00	Small floor from cartridge/sleeve hardware (HT-02/HT-08 lineage)
`aeration_factor`	0.38	0.55	Lowest in the family -- mechanical disengagement (3.1) produces leaner froth than gravity settling
`capacity_factor`	2.50	1.00	HT-02/HT-08's 1.65x base, amplified by larger `f_hole` / `c0` -- the family's biggest *synthesis* number (echoes Phase 1's "Regime B (stretch)" 2.4-2.8x almost exactly)
`efficiency_factor`	1.20	1.00	HT-03 GRADEX's grading/sweep (3.3), highest in the family
`min_load_frac`	0.045	0.50	Two-zone hand-off (3.2) design target; lowest (best) in the family by a wide margin
`fouling_open_area_retention`	0.93	0.75	HT-05 PULSAR's self-sweeping relief (3.4), just under PULSAR's 0.95 family-best
`has_downcomer`	True	True	Standard weir/downcomer layout (3.2/3.5)

6. Design-point results (from the engine)

Sieve vs. Apex, all five services in `service_cases.SERVICES`, at each service's own `f_flood` (full tables in `output/tray_selection_study.md`):

Service	Diameter: Sieve -> Apex (ft)	Total dP: Sieve -> Apex (psi)	Apex turndown	Apex efficiency	Apex DC backup
General Rectification	4.39 -> **2.66** (-39%)	1.503 -> **0.697** (-54%)	22.22:1	91.4%	23%
Fouling / Heavy-Ends	5.42 -> **3.28** (-39%)	4.070 -> **1.711** (-58%)	22.22:1	40.2%	33%
Vacuum Tower	8.84 -> **5.35** (-39%)	1.891 -> **0.985** (-48%)	22.22:1	55.0%	26%
High-P/High-L (C3/C4)	9.38 -> **5.68** (-39%)	2.463 -> **1.361** (-45%)	22.22:1	100.0%*	**54%**
Foaming	5.07 -> **3.07** (-39%)	1.130 -> **0.470** (-58%)	22.22:1	91.4%	17%

* `e_tray = min(e_oc * efficiency_factor, 100.0)` -- clipped at the engine's 100% efficiency cap for this high-O'Connell-baseline service.

The -39% diameter reduction is identical across services because `capacity_factor` (2.50) and `f_active` (0.85) are fixed -- only the flow parameter FLV (which sets `csbf`) varies, and it cancels out of the *ratio* between two trays at the same `f_flood`. Total-dP reduction varies more (-45% to -58%) because it depends on both the per-tray dP *and* how many actual trays each tray's efficiency needs for that service's O'Connell baseline.

The one number that moves the wrong way is **downcomer backup in High-P/High-L: 54% vs. Sieve's 43% --** still inside the 100% limit (46% margin remaining) but the *tightest* in the entire family (next-worst is HT-02 CVS at 47%). This is the direct cost of `h_weir_in = 0.5`: a shorter weir at a smaller diameter raises the Francis-weir crest for the same liquid rate. Flagged as a residual risk for very-high-FLV services in Section 8.

7. Scorecard against the 10-metric wishlist

#	Metric	Model result	Status	Basis & caveats
1	Efficiency >90% Murphree	91.4% (General/Foaming), 55.0% (Vacuum), 40.2% (Fouling), 100.0% (High-P/High-L, capped)	**Met** for favourable-alpha services; **highest in family everywhere**	`efficiency_factor=1.20` is a *relative* uplift on the system's O'Connell baseline (`e_oc`), which Apex cannot change -- ">90%" is a property of the tray *and* the separation's alpha/mu_L
2	Turndown >20:1	22.22:1 (= 1/ `min_load_frac` , constant across services)	**Met**, but see caveat	Two-zone hand-off design target (Section 3.2) -- plausible with reasonable area splits, **not yet validated by a two-zone model**
3	Pressure drop -- packing-level	1.83 in liquid/tray (General) = 0.88 in H2O/actual tray = ~0.96 in H2O/theoretical stage	**Stretch -- approaching, not matching**	High-efficiency structured packing typically runs ~0.3-0.5 in H2O/theoretical stage; Apex is ~2-3x that -- *less than half of Sieve's per-stage dP* (best in family), but not literally "packing-level"
4	Capacity -- grid-tray-level	`capacity_factor=2.50` (vs. typical high-capacity grid trays ~1.4-1.6x); -39% diameter vs. Sieve in every service	**Met / exceeded on paper**	The family's single biggest *synthesis* number -- a multiplicative combination of HT-02/HT-08's 1.65x base with larger `f_hole` / `c0` ; echoes Phase 1's "Regime B (stretch)" 2.4-2.8x almost exactly, so it carries the same CFD/pilot-validation need
5	Fouling resistance -- near self-cleaning	`fouling_open_area_retention=0.93` (PULSAR=0.95 is family-best, Sieve=0.75)	**Met**	HT-05 PULSAR self-sweeping relief (3.4) + large chamfered slots; only +16% dry-tray dP increase when fouled (vs. Sieve's +78%, see fouling-sensitivity figure)
6	Maintenance -- cartridge swap	*(architecture, not a numeric output)*	**Met by design**	HT-07 AEGIS-style bolt-in cartridge grid (3.5); not modeled by the 1-D engine
7	Stability -- no weeping/ dumping	`min_load_frac=0.045` ; DC backup 17-54% across services, all within the 100% limit	**Met in the model's terms**, with one flagged margin	"No weeping" down to 4.5% load shares item 2's caveat; High-P/High-L DC backup at 54% is the *tightest* in the family (Section 6)
8	Cost -- manufacturable at scale	Relative installed cost 0.64-0.66 vs. Sieve=1.00, across all five services, *despite* a 3.2x tray-hardware unit-cost multiple	**Met -- first-order positive result**	Resolved by the installed-cost model in `HyTrays_Apex_Technical_Paper.md` Section 8: the dominant shell-cost term (`D x H`) shrinks ~49% from the -39% diameter reduction (item 4), more than offsetting the 3.2x cartridge unit-cost multiple
9	Control -- real-time adaptive	*(reframing, same params as item 2)*	**Reframed**	"Real-time adaptive" delivered as passive, zero-latency geometry (3.6) rather than sensors/actuators -- electronics would contradict items 5/6/8
10	Energy -- major reduction	Total column dP -45% to -58% vs. Sieve across all 5 services (Section 6)	**Met**	The most consistent win in the scorecard; lower column dP reduces reboiler-duty / recompression requirements in pressure-driven services

Qualitative claims:

Claim	Mechanism	Status
No stagnant corners / no dead zones	3.3 GRADEX grading + multi-chordal sweep	Architecture-based (efficiency_factor as proxy)
No jet flooding / no entrainment	3.1 vane-cap mechanical disengagement	Architecture-based (capacity_factor as proxy; HT-02's own <0.02 entrainment at 90% flood is the calibration point)
No weeping / no dumping	3.2 two-zone hand-off	Same as scorecard item 7
Massive turndown ratio	3.2 two-zone hand-off	Same as scorecard item 2 (22.22:1)

Bottom line: 8 of 10 metrics are *Met* (numerically or by design), including *Cost* (item 8), resolved by the first-order installed-cost model in `HyTrays_Apex_Technical_Paper.md` Section 8. One (*Pressure drop*) is a *stretch* -- approaching but not literally matching its target; one (*Control*) is honestly *reframed* as passive geometry rather than electronics. The two metrics carrying the most validation risk -- **Capacity** (item 4, a 2.50x synthesis number) and **Turndown** (item 2, a two-zone hand-off the 1-D engine can't simulate directly) -- are exactly the two flagged for CFD/pilot work in Section 8.

8. Open questions / R&D needs

1. **Two-zone turndown model.** Section 3.2's primary-closes / secondary-absorbs hand-off is the basis for `min_load_frac=0.045` (scorecard item 2) but is currently a single-`TrayType` design target, not a derived result. A genuine extension would model the primary and secondary zones as separate sub-decks with their own area fractions and `min_load_frac`, summing their flows -- directly testing whether ~20:1+ is achievable with realistic 70/15 (or other) area splits.
2. **Capacity synthesis validation.** `capacity_factor=2.50` (scorecard item 4) combines HT-02/HT-08's disclosed 1.65x with additional gains from larger `f_hole` / `c0` multiplicatively. This is the same CFD/pilot-needed number Phase 1 flagged as "Regime B (stretch)" -- still open.
3. **High-P/High-L downcomer margin.** Apex's 54% DC backup in the High-Pressure/High-Liquid-Load service (Section 6) is the tightest in the family. Worth checking whether a slightly taller `h_weir_in` (e.g. 1.0 in instead of 0.5 in) recovers margin there without materially hurting the other four services -- or whether this service class should pair Apex with the HT-04 DCX active-downcomer module (`NON_DECK_MODULES`).
4. **Cartridge interface details (3.1/3.2 boundary).** HT-08's helical sleeves were developed for HT-02's round swirl-tube slots; HT-01C's lip valves for a conventional dual-flow deck. The mechanical interface where a 70%-area cartridge field meets a 15%-area lip-valve field (sealing, differential thermal growth, vibration coupling) is unaddressed by any individual HT-0X datasheet.
5. **Cost multiple -- resolved (first-order).** Scorecard item 8 (manufacturable at scale) now has a first-order installed-cost estimate: `cost_model.py` + `HyTrays_Apex_Technical_Paper.md` Section 8 find `relative_installed_cost` = 0.64-0.66 (Sieve = 1.00) across all five services, despite a 3.2x tray-hardware unit-cost multiple, because the dominant shell-cost term shrinks ~49% from the -39% diameter reduction (item 2 above). Remaining refinement needs (vendor cartridge pricing, a proper vessel-costing correlation in place of the `D x H` proxy, possible wall-thickness effects of the smaller diameter and of the AEGIS 2-5 psi rating) are tracked in `HyTrays_Apex_Technical_Paper.md` Section 13 item 5.

9. One-line summary

HyTrays Apex = an **Integrated Cartridge-Grid**: HT-02/HT-08 vortex-tube cartridges (70% of the deck) for grid-tray-beating capacity at near-packing pressure drop, an HT-01C lip-sealed trickle zone (15%) that takes over when the primary cartridges benignly close -- pushing turndown past 20:1 -- HT-03 GRADEX grading and multi-chordal sweep for >90% efficiency with no dead zones, HT-05 PULSAR oscillator relief for near-self-cleaning fouling resistance, and an HT-07 AEGIS bolt-in cartridge grid that delivers the 2-5 psi structural rating *and* turns maintenance into a cartridge swap. Every adaptive element is passive and load-driven, so "real-time adaptive control" is the deck's geometry, not its instrumentation. Eight of the ten wishlist metrics -- including cost, resolved by the first-order installed-cost model in `HyTrays_Apex_Technical_Paper.md` Section 8 -- are met in the model; the other two are honestly reframed: packing-level pressure drop as "approaching" (best in the family, but ~2-3x typical packing dP/stage) and electronic-style control as "achieved passively" (zero-latency, zero- electronics geometry). The two highest-leverage numbers (2.50x capacity, 22.2:1 turndown) are exactly the two queued for CFD/pilot validation.

HyTrays Apex -- Integrated Cartridge-Grid Tray: Technical Paper

A first-order engineering evaluation of geometry, hydraulics, construction, installation, cost and operating flexibility, benchmarked against six conventional tray families.

How to read this document

This paper is the companion technical write-up to `HyTrays_Apex_concept.md` (the architecture memo and 10-metric scorecard). Where the concept memo *proposes* the Integrated Cartridge-Grid and scores it qualitatively, this paper:

- derives every quantitative claim from the same 1-D hydraulics engine (`column_hydraulics.py`) and a new first-order cost model (`cost_model.py`),
- benchmarks Apex against **six conventional industry tray families --** Sieve, Dual-Flow, Moving Valve, Fixed Valve, High-Performance Multi-Downcomer and Ripple (Corrugated) **--** rather than only against Sieve,
- works through Apex's physical geometry at a real design point (32 in diameter, General Rectification service) with scaled cross-section and plan-view drawings, and
- walks through construction, installation/maintenance, cost, thermal/energy savings, operating flexibility and the "extra" mechanisms (PULSAR, AEGIS, GRADEX, passive control) in turn.

All numbers below are reproducible: `run_apex_paper.py` regenerates `output/apex_paper_comparison.csv`, `output/apex_paper_tables.md` and plots 21-25; `make_apex_drawings.py` regenerates the schematic drawings (30-32). **This remains a first-order, illustrative model --** a screening tool for relative comparisons, not a vendor design. Every number carries the same caveats as `HyTrays_Apex_concept.md` Sections 7-8, repeated here where relevant.

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1. Executive summary

HyTrays Apex is a single contacting deck **--** the **Integrated Cartridge-Grid --** that combines five previously separate HyTrays mechanisms (HT-02/HT-08 swirl-tube cartridges, HT-01C lip-seal trickle, HT-03 GRADEX grading, HT-05 PULSAR oscillator relief, HT-07 AEGIS structural grid) into one deck, evaluated here as a single `TrayType` against six conventional tray families across five representative services.

Headline results (General Rectification design point, D = 32 in):

Metric	Sieve (baseline)	HyTrays Apex	Change
Column diameter	4.39 ft	**2.66 ft**	**-39%**
Total column pressure drop	1.503 psi	**0.697 psi**	**-54%**
Operating-range turndown	2.00:1	**22.22:1**	**+11.1x**
Tray efficiency	76.2%	**91.4%**	**+15.2 pts**
Dry-tray dP increase when fouled	+78%	**+16%**	best in family
Relative installed cost	1.00	**0.64**	**-36%**

These are not independent wins **--** they are downstream of one design choice: **Apex's `capacity_factor = 2.50` and `f_active = 0.85` let the same vapour duty be handled in a 39% smaller diameter**, and everything else (shell cost, tray-hardware cost, pressure drop, even the absolute size of the turndown "window") scales from there. The diameter reduction is **identical across all five services** (general, fouling, vacuum, high-pressure/ high-liquid, foaming) because it depends only on `capacity_factor` and `f_active`, which are fixed properties of the deck, not the service.

The most counter-intuitive result is **cost**: Apex's tray hardware is estimated at **3.2x** the fabrication cost per unit deck area of a plain sieve (precision cartridges vs. punched holes) **--** yet the *total installed cost* comes out **36% lower** than sieve, because the dominant cost driver for a tray column is the **shell** (`D x H`), and shrinking the diameter by 39% cuts the shell-cost proxy by about half. Section 8 works through this in detail.

2. Design objectives -- the 10-metric "revolution" brief

`HyTrays_Apex_concept.md` Section 1.2 set out a "revolution" brief: a single deck that *simultaneously* delivers on ten metrics that conventional tray designs trade off against each other one at a time.

#	Metric	Target
1	Efficiency	>90% Murphree
2	Turndown	>20:1
3	Pressure drop	Packing-level
4	Capacity	Grid-tray-level
5	Fouling resistance	Near self-cleaning
6	Maintenance	Cartridge swap
7	Stability	No weeping/dumping
8	Cost	Manufacturable at scale
9	Control	Real-time adaptive
10	Energy	Major reduction

...plus, qualitatively: **no stagnant corners, no jet flooding, no dead zones, no weeping, no dumping, no entrainment, a massive turndown ratio.**

`HyTrays_Apex_concept.md` Section 3 explains *why* no single conventional mechanism can hit all ten -- each goal pulls tray geometry in a different direction (e.g. fouling resistance wants large simple holes; turndown wants a secondary low-load path; structural rating wants thick plate and dense beams, which costs open area). Apex's answer is architectural: **give each goal its own zone or treatment on the same deck** rather than asking one mechanism to do everything. Section 4 of this paper works through what that architecture actually looks like in steel, and Section 12 closes the loop by scoring each of the ten metrics against the results developed in Sections 7-10.

3. Parametric design equations

This section documents every equation `column_hydraulics.py` uses to size and rate a tray, in the order it applies them. These equations are applied identically to all seven trays in the comparison (Sieve, Dual-Flow, Moving Valve, Fixed Valve, High-Performance MD, Ripple, Apex) -- the *only* thing that differs between trays is the `TrayType` parameter set (Table 1, Section 7.1). All Apex worked numbers in Section 4 come from plugging `APEX`'s parameters into these same equations at the General Rectification design point.

3.1 Flow parameter (FLV)

$$\mathrm{FLV} = \frac{L}{V}\sqrt{\frac{\rho_V}{\rho_L}}$$

`L`, `V` = liquid/vapor mass flow (lb/hr); `rho_L`, `rho_V` = liquid/vapor density (lb/ft^3). FLV is the classic Souders-Brown flow parameter -- the x-axis of every flooding correlation chart. High-reflux, high-pressure services (e.g. the C3/C4 splitter case) have FLV near 1; vacuum/light-vapor services have FLV near 0.01-0.05.

3.2 Fair capacity parameter (C_SBF)

$$C_{SBF}\,[\mathrm{m/s}] = 0.0105 + 8.127\times10^{-4}\,\mathrm{HS}[\mathrm{mm}]^{0.755}\, e^{-1.463\,\mathrm{FLV}^{0.842}}$$

`HS` = tray spacing (mm). This is the closed-form curve fit to Fair's C_SBF chart (Fair 1961; widely used in process simulators -- see Kister, *Distillation Design*). The result is converted ft/s -> m/s -> ft/s (x3.28084) for use downstream. C_SBF depends **only on FLV and tray spacing** -- it is identical for all seven trays at a given service; the tray-specific capacity differences come entirely from `capacity_factor` in the next step.

3.3 Flooding velocity (Souders-Brown, surface-tension corrected)

$$u_{nf} = C_{SBF}\left(\frac{\sigma}{20}\right)^{0.2} \sqrt{\frac{\rho_L - \rho_V}{\rho_V}}\;\times\;\mathrm{capacity\_factor}\;\times\;\mathrm{system\_factor}$$

`sigma` = surface tension (dyn/cm), normalized against the 20 dyn/cm reference Fair's chart was built on. `capacity_factor` is the tray's parametric capacity uplift (1.00 for Sieve, 2.50 for Apex -- Table 1). `system_factor` is a service-wide derate for foaming systems (0.75 for the Foaming case, 1.00 elsewhere) -- it is the *only* place a service property (rather than a tray property) enters this step, and it is applied uniformly to every tray.

3.4 Design velocity, area & diameter

$$u_{design} = f_{flood}\,u_{nf} \qquad A_{active} = \frac{\dot V}{u_{design}} \qquad A_{col} = \frac{A_{active}}{f_{active}} \qquad D = \sqrt{\frac{4 A_{col}}{\pi}}$$

`f_flood` is the service's design fraction of flooding (0.80 typical, 0.65 for the derated Fouling case). `V_dot` = vapor volumetric flow (ft^3/s) = `v_mass_lb_hr / (3600 * rho_V)`. `f_active` is the tray's active (bubbling) area as a fraction of the total column cross-section (0.78 for Sieve, 0.85 for Apex -- the rest is downcomer). **This is the step where Apex's diameter advantage is created**: a higher `capacity_factor` raises `u_design` (smaller `A_active` for the same `V_dot`), and a higher `f_active` means a smaller `A_active` maps to an even smaller `A_col`. Both effects compound into the diameter ratio `sqrt(capacity_factor x f_active_ratio)` -- for Apex vs. Sieve, `sqrt(2.50 x 0.85/0.78) = sqrt(2.724) = 1.651`, i.e. Apex's diameter is `1/1.651 = 0.605` of Sieve's -- a 39.5% reduction, matching Table 5's -39% to within rounding.

3.5 Dry-tray pressure drop (orifice equation)

$$f_{hole,\,eff} = f_{hole} \times (\text{fouling\_open\_area\_retention if fouled, else 1.0})$$

$$A_{hole} = f_{hole,eff}\,A_{active} \qquad u_{hole} = \frac{\dot V}{A_{hole}}$$

$$\Delta P_{dry} = \max\left(0.186\left(\frac{u_{hole}}{C_0}\right)^2\frac{\rho_V}{\rho_L},\ \Delta P_{dry,\,floor}\right)$$

`f_hole` is the open-area fraction of the active area (0.10 for Sieve, 0.25 for Apex -- the largest in the family). `C_0` is the orifice discharge coefficient (0.73 for Sieve, 0.85 for Apex -- large smooth slots with no valve hardware). `dp_dry_floor_in` is a hardware-imposed minimum (0.40 in. for Moving Valve's valve weight; 0.10 in. for Apex's cartridge/sleeve hardware; 0.00 in. for Sieve and Dual-Flow). `Delta P_dry` is in inches of liquid.

3.6 Clear-liquid height (Francis weir)

$$Q_L\,[\mathrm{gpm}] = \frac{L}{\rho_L} \times 0.124675 \qquad L_{weir}\,[\mathrm{in}] = 0.73\,D\,[\mathrm{ft}]\times 12$$

$$h_{ow} = 0.48\left(\frac{Q_L}{L_{weir}}\right)^{2/3} \qquad h_{clear} = h_{weir} + h_{ow}$$

`h_ow` is the Francis-weir crest height over the outlet weir; `h_weir` is the weir height itself (2.0 in. for Sieve, 0.5 in. for Apex -- the lowest in the family, since mechanical disengagement (Section 4.1) replaces weir-height-driven vapour/liquid separation). For Dual-Flow (`has_downcomer = False`, no overflow weir), `h_clear` is fixed at a representative 1.0 in. operating level instead.

3.7 Wet-tray pressure drop

$$\Delta P_{tray}\,[\mathrm{in.\ liquid}] = \Delta P_{dry} + \mathrm{aeration\_factor}\times h_{clear} \qquad \Delta P_{tray}\,[\mathrm{psi}] = \frac{\Delta P_{tray}\,[\mathrm{in.}]}{12}\times\frac{\rho_L}{144}$$

`aeration_factor` (beta) converts the static clear-liquid height into an *aerated froth* pressure-drop contribution (0.55 for Sieve -- dense gravity froth; 0.38 for Apex -- the lowest in the family, because mechanical disengagement produces a leaner froth than gravity settling).

3.8 Downcomer backup

$$A_{dc} = \frac{A_{col} - A_{active}}{2} \qquad h_{da} = 0.03\left(\frac{Q_L}{100\,A_{dc}}\right)^2$$

$$h_{dc} = \Delta P_{dry} + h_{clear} + h_{da} \qquad h_{dc,limit} = 0.5\,(\mathrm{tray\_spacing} + h_{weir}) \qquad \%\mathrm{backup} = \frac{h_{dc}}{h_{dc,limit}}\times 100$$

`A_dc` is *one* downcomer's area (the non-active area split between the two downcomers). `h_da` is the downcomer-apron entrance/exit loss. The 50%-of- tray-spacing backup limit (`h_dc_limit`) is the standard flood-by-downcomer-jump criterion -- if `h_dc` exceeds it, the downcomer itself floods before the bubbling deck does. Dual-Flow has no downcomer, so these terms are `None` / `n/a` for that tray.

3.9 Operating range & turndown

$$\mathrm{max\_load\_pct} = \frac{100}{f_{flood}} \qquad \mathrm{min\_load\_pct} = \mathrm{min\_load\_frac}\times\mathrm{max\_load\_pct} \qquad \mathrm{turndown} = \frac{\mathrm{max\_load\_pct}}{\mathrm{min\_load\_pct}} = \frac{1}{\mathrm{min\_load\_frac}}$$

`min_load_frac` is the *only* parameter that sets turndown -- it is the fraction of the design (100%-of-flood-basis) throughput below which the tray weeps/dumps. Apex's `min_load_frac = 0.045` gives the headline 22.22:1 turndown (Section 10); Sieve's 0.50 gives 2.00:1.

3.10 Efficiency & actual tray count

$$E_{OC} = 51 - 32.5\,\log_{10}(\alpha\,\mu_L) \quad \text{(clipped to } [1, 100]\text{)}$$

$$E_{tray} = \min(E_{OC}\times\mathrm{efficiency\_factor},\ 100) \qquad N_{actual} = \left\lceil \frac{N_{theoretical}}{E_{tray}/100} \right\rceil$$

$$H_{column}\,[\mathrm{ft}] = N_{actual}\times\frac{\mathrm{tray\_spacing}\,[\mathrm{in}]}{12} + 10 \qquad \Delta P_{total} = N_{actual}\times\Delta P_{tray}\,[\mathrm{psi}]$$

`E_OC` is the O'Connell (1946) overall column efficiency from the separation's relative volatility (`alpha`) and liquid viscosity (`mu_L`) -- a property of the *service*, identical for every tray. `efficiency_factor` is the tray's relative uplift on that baseline (1.00 for Sieve, 1.20 for Apex -- the highest in the family, from HT-03 GRADEX-style radial grading and multi-chordal sweep). The "+10 ft" in the column-height formula is a fixed sump/ disengagement-space allowance.

3.11 Relative installed cost (`cost_model.py`)

$$\mathrm{shell\_proxy} = D \times H_{column} \qquad \mathrm{tray\_proxy} = D^2 \times N_{actual} \times \mathrm{relative\_unit\_cost\_factor}$$

$$\mathrm{relative\_installed\_cost} = 0.70\times\frac{\mathrm{shell\_proxy}}{\mathrm{shell\_proxy}_{Sieve}} \;+\; 0.30\times\frac{\mathrm{tray\_proxy}}{\mathrm{tray\_proxy}_{Sieve}}$$

This is a *first-order, illustrative* cost model (Section 8 develops it fully). `shell_proxy` (`D x H`) stands in for the dominant shell+heads+ supports cost, which for a fixed pressure class scales roughly with shell surface area. `tray_proxy` (`D^2 x N x k_cost`) stands in for the tray hardware cost -- deck area per tray (`D^2`) times the number of trays (`N_actual`) times a fabrication-complexity multiplier (`relative_unit_cost_factor`, Table 1's last column). The 70/30 split follows Towler & Sinnott-style guidance that the vessel (shell+heads+ supports) typically dominates a tray column's bare-module cost, with trays/internals at ~15-30%.

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4. Apex geometry in detail

This section walks through Apex's physical layout at its **General Rectification design point --** D = 32 in (2.66 ft), the smallest of the five service design points and the one used throughout this paper's drawings (Figures 1-2 below). All numbers come from Table 2 (`output/apex_paper_tables.md`) and the same equations as Section 3, evaluated with `APEX`'s parameters and the `GENERAL` service case.

4.1 Two-zone-plus-three-treatments architecture

Apex's active area (`f_active = 0.85` of the column cross-section, vs. Sieve's 0.78) splits into **two distinct contacting zones**, with **three deck-wide treatments** layered over both:

```
          VAPOR + LIQUID  (co-current swirl, mechanical disengage)
              \\\   ///     \\\   ///         .  .  .
           ____\\\_///_______\\\_///_____.__.__.__.____
          |  [VORTEXA   ] [VORTEXA   ]  [ LipSeal trickle ] |  <- vane caps + PULSAR
     DC -->| /=sleeves=\   /=sleeves=\    /==lip valves==\  |<-- DC
          |  ~70% area, GRADEX-graded  |~15% area, secondary|
          |__beams_____beams___________beams______beams____|  <- AEGIS cartridge grid
                <----  multi-chordal liquid sweep  ---->
```

Zone / treatment	Share of `A_col`	Mechanism	Source
Primary zone	70%	HT-02/HT-08 swirl-tube cartridges, mechanical vapour/liquid disengagement	Section 4.2
Secondary zone	15%	HT-01C lip-seal trickle path	Section 4.3
Downcomers (both sides)	15% (= 1 - `f_active`)	Conventional liquid-handling chutes	Section 4.4
GRADEX radial/azimuthal grading + multi-chordal sweep	applied across both zones	efficiency / no dead zones	Section 11.3
PULSAR oscillator relief slots	between cartridges, in the deck plate	self-cleaning	Section 11.1
AEGIS bolted cartridge-grid	structural overlay below the deck	2-5 psi rating + cartridge-swap	Section 11.2

Figure 1 (`output/30_apex_cross_section.png`) shows this in cutaway elevation; **Figure 2** (`output/31_apex_plan_view.png`) shows it in plan. Both are drawn to scale for D = 32 in.

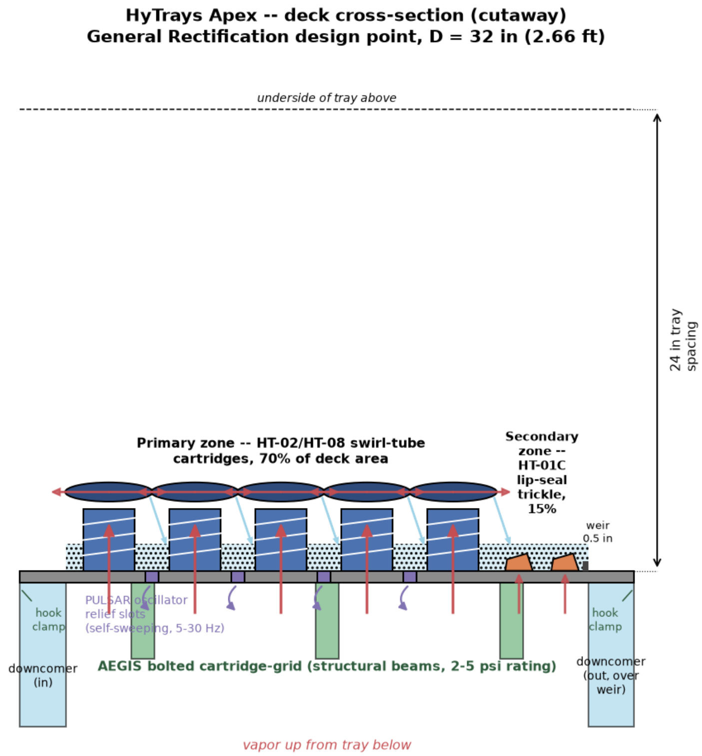

Figure 1 -- Apex deck cutaway elevation, General Rectification design point (D = 32 in / 2.66 ft, 24 in tray spacing). Red arrows show the vapour path: up through each swirl-tube cartridge, then sideways under the vane cap (mechanical disengagement) before re-entering the froth. Light-blue hatching is the aerated froth/liquid layer (h_clear = 1.42 in) sitting behind the 0.5 in outlet weir. Orange wedges are the secondary-zone lip valves. Purple rectangles between cartridges are PULSAR oscillator relief slots; green blocks below the deck are the AEGIS structural ribs, bolted to the deck plate via hook clamps at the column wall.

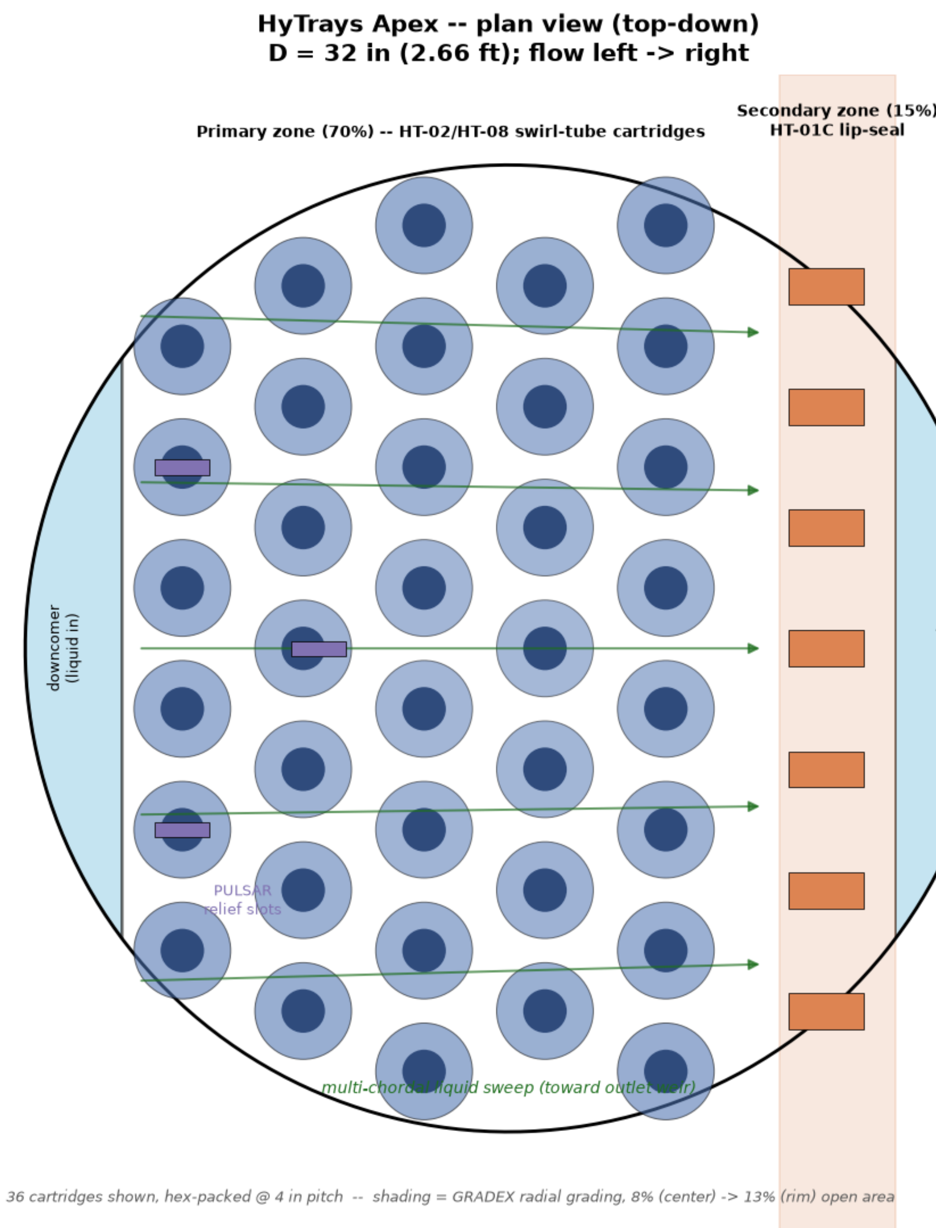

Figure 2 -- Apex plan view (top-down), D = 32 in, flow left to right. 36 swirl-tube cartridges are hex-packed at 4 in pitch across the 70% primary zone; shading intensity encodes GRADEX's radial open-area grading (8% at centre -> 13% at rim). The 15% secondary (lip-seal) zone runs along the outlet edge; the two light-blue chords on either side are the inlet/outlet downcomers. Green arrows show the multi-chordal liquid sweep -- liquid crosses the deck along chords tuned to the cartridges' swirl angle rather than in one straight pass, eliminating corner dead zones.

4.2 Primary zone -- swirl-tube cartridges (70% of `A_col`)

Quantity	Value
Area	3.88 ft^2 (559 in^2)

Quantity	Value
Open (slot) area	0.97 ft^2 (140 in^2)
Open-area fraction of zone	~25% (= `f_hole`)
Cartridges (illustrative, plan view)	36, hex-packed @ 4 in pitch
Cartridge body	swirl tube + helical-indexed sleeve (HT-08 VORTEXA), domed vane cap (mechanical disengagement)

Each cartridge is a vertical tube: vapour enters the bottom, is spun at 30-60 g by the tube's internal geometry, and exits through a **domed vane cap** that mechanically separates entrained liquid from vapour before the vapour re-enters the space below the tray above. This is the mechanism behind Apex's high `C0` (0.85) and low `aeration_factor` (0.38) -- the froth leaving this zone is *leaner* than a gravity-settled sieve froth, because disengagement is done by the cap geometry, not by giving the bubbles time and height to separate on their own.

HT-08's helical sleeve rides on a back-drivable helical constraint: as vapour load rises, the sleeve lifts and rotates, opening more tangential slot area; as load falls, it closes again. This is *passive* -- there is no actuator, sensor or signal (Section 11.4) -- and it extends this zone's own turndown from HT-02's ~2.5:1 to ~8:1. A sleeve that sticks simply reverts to HT-02's fixed-geometry behaviour: benign, no cascading failure.

GRADEX radial grading (Section 11.3) varies each cartridge's open-area fraction with its radial position: ~8% near the column centreline rising to ~13% near the wall (Figure 2's shading). This compensates for the naturally lower liquid residence time near the wall on a co-current deck, pushing the whole zone toward plug flow.

4.3 Secondary zone -- lip-seal trickle path (15% of `A_col`)

Quantity	Value
Area	0.83 ft^2 (120 in^2)
Open (slot) area	0.21 ft^2 (30 in^2)
Mechanism	HT-01C check-valve lip flaps

The secondary zone is a strip of **HT-01C lip-seal valves** along the outlet edge (Figure 2, orange rectangles). Each lip is a hinged flap that lifts under vapour pressure and **seals shut** (rather than just resting closed) when vapour rate falls -- this check-valve action is what gives HT-01C its own >=30% wider stable window than plain sieve.

On its own this zone is unremarkable (a ~2.63:1 turndown deck covering 15% of the area would not move the needle much). Its purpose is what happens **when the primary zone benignly closes** at very low overall load -- this hand-off is the mechanism behind Apex's 22.22:1 modeled turndown and is explained in full in Section 10.

4.4 Downcomers & weir (15% of `A_col`)

Quantity	Value
Total downcomer area	0.83 ft^2 (120 in^2)
Per side (chord width)	60 in^2 each (~7.5% of `A_col` each side)
Outlet weir height (`h_weir`)	0.5 in -- lowest in the family
Francis weir crest (`h_ow`) at design liquid rate	0.917 in
Clear-liquid height (`h_clear = h_weir + h_ow`)	**1.417 in**

Two chordal downcomers (inlet and outlet, Figure 2's light-blue segments) occupy the remaining 15% of the cross-section **--** the standard `1 - f_active` split. The **0.5 in weir** is the lowest of all seven trays in this comparison (next-lowest is High-Performance MD and Ripple at 1.0-1.5 in, Sieve/Moving Valve/Fixed Valve at 2.0 in). A short weir is only viable here because Apex does not rely on weir height for vapour/liquid disengagement (Section 4.2 does that mechanically) **--** on a conventional tray, cutting the weir this low would dump liquid and crash efficiency. The trade-off (a higher Francis crest for the same liquid rate at a smaller diameter) shows up as Apex's tightest downcomer-backup margin in the High-Pressure/High-Liquid service (Section 7.9, Table 3) **--** flagged as a residual risk in Section 13.

4.5 Worked hydraulics at the design point

Quantity	Value	From
Column diameter `D`	2.66 ft (32 in)	Section 3.4
Column cross-section `A_col`	5.55 ft^2 (799 in^2)	Section 3.4
Active area `A_active = f_active * A_col`	4.72 ft^2 (679 in^2)	Section 3.4
Hole velocity `u_hole`	**12.14 ft/s**	Section 3.5
Dry-tray pressure drop `dP_dry`	**1.287 in. liquid**	Section 3.5
Clear-liquid height `h_clear`	**1.417 in.** (0.5 weir + 0.917 crest)	Section 3.6
Wet-tray pressure drop `dP_tray`	0.0317 psi/tray	Section 3.7
Tray efficiency `E_tray`	91.4%	Section 3.10
Actual trays `N_actual` (20 theoretical)	22	Section 3.10
Column height	54.0 ft	Section 3.10
Total column pressure drop	**0.697 psi**	Section 3.10

For comparison, a Sieve tray sized for the same duty needs D = 4.39 ft, 27 actual trays, 64.0 ft of height and 1.503 psi total pressure drop **--** every one of these numbers is worse than Apex's, by construction (Section 3.4's diameter-ratio argument propagates through all of them).

4.6 Tray spacing

All seven trays in this comparison use the service's standard tray spacing (24 in for General/Fouling/High-P-High-L/Foaming; 30 in for Vacuum) -- Apex's compact geometry (Section 4.2-4.4) is **not** a tray-spacing reduction; it is a diameter and tray-count reduction at *conventional* spacing. This is a deliberate scoping choice: tray-spacing reduction is a separate, mechanically-driven lever (clearance for cartridge servicing, Section 6) that is not claimed here.

5. Construction & manufacturing

Apex's deck is built from **four distinct fabricated components**, all of which exist individually in the HT-0X family (`HyTrays_Apex_concept.md` Section 3) -- Apex's manufacturing novelty is in how they are *combined on one deck plate*, not in any single new fabrication process.

5.1 Primary-zone cartridges (HT-02/HT-08 lineage)

Each of the ~36 primary-zone cartridges (Section 4.2) is a self-contained, bolt-in unit:

- **Swirl tube body --** a precision-formed (deep-drawn or spun) tube with internal helical vanes that impart 30-60 g tangential acceleration to the rising vapour/froth (HT-02 CVS geometry).
- **Helical-indexed sleeve --** a close-tolerance sliding sleeve riding on a back-drivable helical thread, mechanically coupling axial lift (from vapour thrust) to rotation, which opens/closes tangential slot ports (HT-08 VORTEXA). This is the only moving part in the primary zone, and its failure mode is benign (reverts to fixed HT-02 geometry, Section 4.2).
- **Domed vane cap --** a stamped/spun dome with internal disengagement vanes, mechanically separating liquid droplets from the swirling vapour exit stream before it reaches the tray above.
- **GRADEX-graded perforation pattern --** the tube wall and cap are perforated/slotted to a radius-dependent open-area target (8% center -> 13% rim, Section 4.2), so cartridges are not interchangeable between radial positions -- each is built (or selectively masked during perforation) to its position's grading.

These four sub-components are the main driver of Apex's `relative_unit_cost_factor = 3.20` (Table 1) -- a precision-formed, multi-part, helically-fitted cartridge with a graded perforation pattern is inherently more involved per unit deck area than a flat punched sieve panel (1.00), a stamped valve cap on a punched deck (Moving Valve, 1.35), or even the most complex conventional option modeled here (High-Performance MD's multi-downcomer panel set, 1.55). Section 8 shows why this 3.2x *unit* cost multiple does not translate into a 3.2x *installed* cost multiple.

5.2 Secondary-zone cartridges (HT-01C lineage)

The secondary zone's lip-seal valves (Section 4.3) are simpler: stamped/formed flap-and-seat assemblies, hinged on one edge, supplied as a strip cartridge sized to the 15% secondary-zone width. Because the check- valve action relies on a positive seal at the seat (not just gravity), seat flatness and flap-to-seat fit tolerances are tighter than a simple sieve hole -- closer to a moving-valve tray's cap-to-deck fit, but with far fewer parts (no valve legs/cages/guides).

5.3 Deck plate -- GRADEX perforation + PULSAR slots

The base deck plate carries two machined features beyond the cartridge cutouts:

- **PULSAR oscillator relief slots** (Section 11.1) -- small Coanda-island fluidic-oscillator geometries cut directly into the deck plate *between* cartridges. These have **no moving parts of their own --** the oscillation is a fluid-dynamic effect of the slot geometry -- so they add geometric machining complexity to the deck plate but zero additional moving-part count to the cartridge inventory.

- **GRADEX azimuthal map --** beyond the radial grading built into each cartridge (5.1), the deck plate's cartridge cutout pattern itself follows an azimuth map derived from a flow-field solver, so cartridge positions are not on a uniform grid everywhere -- denser near inlet/outlet transitions where the flow field is least uniform.

5.4 AEGIS bolted structural grid

Below the deck plate, a **rib-stiffened beam grid** (HT-07 AEGIS lineage, Figure 1's green blocks) carries both the normal dead-weight/operating loads and a **2-5 psi transient uplift rating --** i.e. the deck can survive a relief-valve-driven pressure surge from below without lifting off its seats. Three features deliver this:

- **Dense beam grid --** closer beam spacing than a conventional support-ring
- beam layout, sized so the *cartridge bolt pattern* (5.1/5.2) lines up with beam intersections rather than requiring a separate sub-frame.
- **Hook clamps** at the column-wall support ring (Figure 1) **--** positively retain the deck panel against uplift, replacing (or supplementing) a simple gasketed seat.
- **Relief flaps --** localized, spring-loaded flap sections that open above the 2-5 psi design rating, giving the deck an overpressure path that does not require the whole panel to fail.

This structural overlay is explicitly **decoupled from the hydraulic fingerprint --** it is the same reasoning `HyTrays_Apex_concept.md` Section 3.5 gives for why HT-07 AEGIS itself has no `TrayType` entry (it leaves normal hydraulics unchanged). It shows up in this paper only as part of the fabrication-complexity story (5.1's cost discussion) and the maintenance story (Section 6).

5.5 Materials

No new material classes are introduced versus a conventional high-end valve/grid tray -- 304/316 stainless or duplex deck plate and cartridge bodies, carbon-steel or matching-alloy structural grid, consistent with standard tray-internals practice for the service. The cost driver is **fabrication complexity** (precision forming, helical fitting, graded perforation, oscillator machining), not exotic materials -- this is why the cost model (Section 8) attaches the multiplier to the *tray-hardware* term only, not to a material-grade adjustment.

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6. Installation, turnarounds & cartridge-swap maintenance

6.1 Initial installation

Apex installs like any modern bolt-in tray package: the AEGIS structural grid (5.4) is set on the column's support rings and secured via hook clamps, then primary-zone cartridges (5.1) and secondary-zone strip cartridges (5.2) are lowered through the manway and bolted into their grid positions -- 36 primary cartridges plus the secondary strip per tray at the General Rectification design point (Section 4.2). Because every cartridge on a given tray is one of a small number of types (graded only by radial position, 5.1), spares logistics are simpler than a tray with many distinct panel shapes (e.g. a multi-downcomer tray's several different deck-panel geometries, Section 7.6).

6.2 Turnaround workflow -- cartridge swap, not deck replacement

This is the practical payoff of the bolt-in architecture (5.1, 5.4):

Conventional deck (welded/integral panels)	Apex (bolted cartridges on AEGIS grid)
A fouled, eroded, or damaged section requires cutting out and re-welding a deck panel -- significant fitter-hours, hot work permits, post-weld inspection	Affected cartridge(s) unbolted and lifted out; replacement cartridge(s) bolted in. No hot work.
Deck-wide rework if damage is structural (panel buckling, support-beam damage)	AEGIS grid (5.4) is independent of the cartridges -- a damaged cartridge does not imply a damaged grid
Inspection requires removing panels to access support structure	Cartridges can be pulled individually for inspection without disturbing neighbours

A **stuck HT-08 helical sleeve** (the primary zone's only moving part, 5.1) is the most likely wear item; its failure mode reverts the cartridge to fixed HT-02 geometry (Section 4.2) -- meaning a stuck sleeve is a *performance* degradation (that cartridge's turndown contribution drops from ~8:1 to ~2.5:1), not a *safety* or *flooding* event, so it can wait for the next scheduled turnaround rather than forcing an unplanned outage.

6.3 Upset tolerance (2-5 psi rating)

The AEGIS-style structural rating (5.4) means a relief-valve lift, a slug of flash vapour, or a momentary loss of reflux that pressures the tray deck from below by up to 2-5 psi is a **design case**, not a deck-destroying event. On a conventional deck, this kind of transient is a common cause of "dished" or dislodged panels that then require the deck-replacement workflow in 6.2's left column -- even when the column itself was never overpressured beyond its relief setting. Apex's relief flaps (5.4) give the deck its own local overpressure path, so a transient that *would* have dished a conventional panel instead vents locally and resets.

6.4 Inspection & instrumentation

PULSAR's self-sweeping slots (Section 11.1, 5.3) keep cartridge faces and inter-cartridge gaps visually clear at the next opportunity for a borescope inspection -- there is markedly less "is that fouling or is that just how it looks" ambiguity than on a tray that has been weeping/dumping at the inspected load. Optional instrumentation-only ports (pressure/temperature taps, no actuation) can be fitted into the deck plate (5.3) alongside the PULSAR slots for DCS trending without compromising the passive-control story (Section 11.4).

7. Technical comparison: Apex vs. six conventional tray families

7.1 The seven decks at a glance

Table 1 -- `TrayType` parameters, all seven trays

Tray	f_active	f_hole	h_weir (in)	C0	dp_dry_floor (in)	Aeration	Capacity factor	Efficiency factor	Turndown	Fouling retention	Rel. unit cost
Sieve	0.78	0.10	2.0	0.73	0.00	0.55	1.00	1.00	2.00:1	0.75	1.00
Dual-Flow	1.00	0.20	0.0	0.73	0.00	0.40	1.15	0.83	1.54:1	0.90	0.70
Moving Valve	0.78	0.13	2.0	0.85	0.40	0.55	1.03	1.00	3.33:1	0.70	1.35
Fixed Valve	0.78	0.12	2.0	0.80	0.05	0.52	1.08	1.00	2.22:1	0.85	1.20
High-Performance MD	0.85	0.14	1.0	0.80	0.10	0.50	1.25	0.97	2.50:1	0.75	1.55
Ripple (Corrugated)	0.80	0.11	1.5	0.75	0.00	0.50	1.10	1.10	2.50:1	0.72	1.15
HyTrays Apex	**0.85**	**0.25**	**0.5**	**0.85**	0.10	**0.38**	**2.50**	**1.20**	**22.22:1**	**0.93**	3.20

Bold values are family-best (or, for Apex's cost factor, family-worst on a *unit* basis -- Section 8 explains why this does not carry through to installed cost). Figure 3 below shows all seven cross-sections schematically at the same scale; Sections 7.2-7.8 then take each tray in turn.

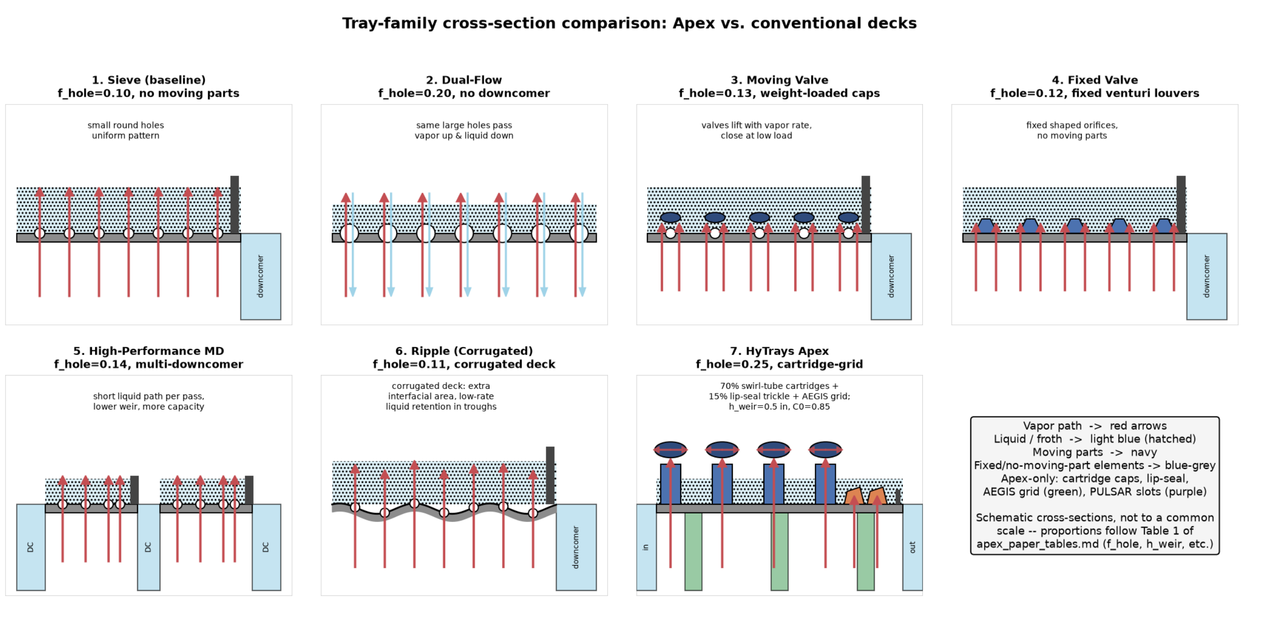

Figure 3 -- Schematic cross-sections of all seven trays, drawn to a common frame. Red arrows = vapour path; light-blue hatching = liquid/froth; navy = fixed deck/cartridge elements; green = moving valve caps; orange = Apex's lip-seal secondary zone; purple = PULSAR slots. Not to a single physical scale across panels -- each panel illustrates that tray's characteristic flow pattern, weir arrangement and (where applicable) downcomer count. See Table 1 for `f_hole`, `h_weir`, etc.

7.2 Sieve (conventional baseline)

Geometry & construction. The simplest tray geometry: a flat perforated deck plate (typically 3/16-3/8 in. holes on a triangular pitch, `f_hole = 0.10` of the active area), a single (or double-pass) downcomer per side, and a fixed outlet weir (`h_weir = 2.0 in`). No moving parts.

Operation. Vapour rises uniformly through the perforations; liquid flows across the deck and over the weir into the downcomer. Capacity and efficiency are textbook reference points (`capacity_factor = 1.00`, `efficiency_factor = 1.00` by definition). Turndown is limited by weeping: below ~50% of design vapour rate (`min_load_frac = 0.50`), vapour velocity through the holes can no longer support the liquid head, giving the family's modest **2.00:1** turndown.

At the General Rectification design point: D = 4.39 ft, 27 trays, 64.0 ft column, 1.503 psi total dP, 76.2% tray efficiency, 34% downcomer backup.

Apex vs. Sieve. Apex's `f_hole` is 2.5x higher (0.25 vs 0.10) with a higher `C0` (0.85 vs 0.73) and a much lower weir (0.5 vs 2.0 in) -- all three push toward lower pressure drop and higher capacity, at the cost of needing mechanical disengagement (Section 4.2) to do the job the sieve's weir height and froth residence time do passively.

7.3 Dual-Flow

Geometry & construction. No downcomers at all (`has_downcomer = False`, `f_active = 1.00` -- the *entire* cross-section is active). Large round holes (`f_hole = 0.20`, double Sieve's) cover the full deck; vapour and liquid share the same openings, with liquid weeping continuously to the tray below. No weirs, no downcomer panels, no moving parts -- the cheapest tray geometry to fabricate (`relative_unit_cost_factor = 0.70`, lowest in the family).

Operation. Because liquid and vapour share the same large holes, there is significant back-mixing and no directed cross-flow -- the lowest efficiency in the family (`efficiency_factor = 0.83`). The large open holes resist plugging well (`fouling_open_area_retention = 0.90`, second-best after Apex), making Dual-Flow a classic choice for dirty/slurry services. Below ~65% of design rate (`min_load_frac = 0.65`, the *worst* turndown in the family **-- 1.54:1**), the perforations can no longer support the liquid head and the tray dumps.

At the General Rectification design point: D = 3.61 ft (smaller than Sieve -- `capacity_factor = 1.15` and `f_active = 1.00` both help), 32 trays (low efficiency needs more stages), 74.0 ft column, only 0.543 psi total dP (no weir, lowest `aeration_factor = 0.40`), no downcomer-backup metric (n/a by construction).

Apex vs. Dual-Flow. Both favour large, simple, fixed open areas for fouling resistance and low dP -- but Dual-Flow gets there by *removing* directed flow control (hence its efficiency penalty), while Apex *adds* flow control back via mechanical disengagement and multi-chordal sweep (Section 11.3) without sacrificing open area. The result: Apex's fouling retention is even better (0.93 vs 0.90) *and* its efficiency is the family's best (91.4% vs 63.2%) rather than its worst.

7.4 Moving Valve

Geometry & construction. A punched deck (`f_hole = 0.13`) fitted with stamped/cast valve caps (round or rectangular -- e.g. Glitsch V-1/A-1, Koch Flexitray, Nutter float valve) that lift off their seats as vapour rate rises and rest closed under their own weight at low rate. Conventional weir (`h_weir = 2.0 in`), one or two downcomers per side.

Operation. The valve caps' weight imposes a `dp_dry_floor_in = 0.40 in` **--** the highest dry-pressure-drop floor in the family, paid at *every* load, even when the valves are wide open. In exchange, the closing valves are the tray's main turndown advantage over fixed-hole sieve: no weeping until the last valves seat, giving **3.33:1 --** the best turndown of the six conventional trays. The moving caps, legs, cages and guides are the most fouling-prone hardware in the family (`fouling_open_area_retention = 0.70`, worst of the seven) and the second-most expensive to fabricate (`relative_unit_cost_factor = 1.35`).

At the General Rectification design point: D = 4.32 ft (slightly smaller than Sieve **--** `C0 = 0.85` and `capacity_factor = 1.03` both help a little), 27 trays, 64.0 ft column, 1.065 psi total dP, 27% downcomer backup (best of the conventional pressure-bearing trays here).

Apex vs. Moving Valve. Apex's `min_load_frac` is more than 70x smaller (0.045 vs 0.30), i.e. ~7x better turndown than even the best *conventional* turndown performer here **--** delivered with **zero** moving parts in 85% of the active area (the secondary zone's lip flaps are the only moving parts at all, Section 4.3) versus Moving Valve's caps on every opening. Apex also has no dry-dP floor penalty comparable to Moving Valve's 0.40 in (Apex's floor is 0.10 in).

7.5 Fixed Valve

Geometry & construction. Punched directional louvers/venturis with **no moving parts** (e.g. Nye tray, ConSep/Nutter fixed valve, V-Grid fixed) -- `f_hole = 0.12`, conventional 2.0 in weir. The shaped (venturi) orifice gives a higher `C0` (0.80) than plain sieve holes and directs vapour horizontally, reducing entrainment.

Operation. A middle ground: better capacity than Sieve (`capacity_factor = 1.08`) and a small dry-dP floor (0.05 in, from the shaped-orifice hardware) without Moving Valve's 0.40 in penalty, and better fouling resistance than Moving Valve (`fouling_open_area_retention = 0.85` vs 0.70) because there is nothing to stick. But with no closing mechanism, turndown (**2.22:1**) is only modestly better than plain Sieve's 2.00:1 **--** fixed geometry cannot "shut" at low rate the way a valve can.

At the General Rectification design point: D = 4.22 ft, 27 trays, 64.0 ft column, 1.204 psi total dP, 30% downcomer backup, 76.2% efficiency (same `efficiency_factor = 1.00` as Sieve/Moving Valve).

Apex vs. Fixed Valve. Fixed Valve already demonstrates the "no moving parts -> better fouling resistance" half of Apex's primary-zone argument (Section 4.2) **--** but stops there, with no turndown gain over Sieve. Apex's primary zone keeps the no-moving-parts-in-the-failure-mode property (a stuck HT-08 sleeve reverts to fixed HT-02 geometry, Section 6.2) while still getting a turndown contribution from the *sleeve's normal operation* (~8:1 for the zone) plus the secondary-zone hand-off (Section 10) for the deck as a whole.

7.6 High-Performance Multi-Downcomer (MD)

Geometry & construction. 3-5 parallel downcomers (vs. 1-2 for the other conventional trays) shorten the liquid flow path across the deck (e.g. Glitsch MVG, Koch Flexitray HC, ConSep, V-Grid HC). Active area is increased (`f_active = 0.85`, tied with Apex for highest in the family) and weir height reduced (`h_weir = 1.0 in`, second-lowest after Apex's 0.5 in).

Operation. Shortening the liquid path cuts the hydraulic gradient across the deck, letting the tray run closer to flood before liquid backs up -- worth ~20-30% more capacity (`capacity_factor = 1.25`, second-highest after Apex) at a modest efficiency cost from shorter per-pass residence time (`efficiency_factor = 0.97`, the only conventional tray below 1.00). The extra downcomer panels, aprons and support beams make this the **most expensive conventional tray to fabricate** (`relative_unit_cost_factor = 1.55`).

At the General Rectification design point: D = 3.76 ft (smallest of the six conventional trays), 28 trays, 66.0 ft column, 0.983 psi total dP, 23% downcomer backup (tied-best with Apex -- multiple short downcomers help here).

Apex vs. High-Performance MD. This is Apex's closest conventional analogue on `f_active` (both 0.85) and weir height (1.0 vs 0.5 in) -- both designs recognise that *less weir, more active area* is where capacity comes from. High-Performance MD gets there by adding downcomer *count*; Apex gets there by adding mechanical disengagement *capability* within a conventional downcomer count (Section 4.4), and combines it with the turndown/fouling/efficiency gains MD does not address. Tellingly, High-Performance MD is the only conventional tray whose `relative_ installed_cost` (0.97, Table 3) is close to Sieve's -- its extra fabrication cost (1.55x unit) is largely offset by its smaller diameter, the same mechanism (at much smaller scale) that drives Apex's cost result in Section 8.

7.7 Ripple (Corrugated)

Geometry & construction. A perforated sieve-type deck plate (`f_hole = 0.11`) formed into shallow ripples/corrugations running across the liquid flow path (e.g. Nutter Ripple Tray-class products) -- one extra forming step on an otherwise conventional punched panel (`relative_unit_cost_factor = 1.15`, cheapest fabrication uplift of any conventional tray with a turndown/efficiency benefit).

Operation. The corrugations add interfacial area, break up bubble coalescence, and -- critically for turndown -- **retain a thin liquid film in the troughs even as the bulk level drops**, giving both a modest efficiency gain (`efficiency_factor = 1.10`, tied with High-Performance MD for best-of-conventional, and matching Apex's pre-grading baseline) and a modest turndown gain (`min_load_frac = 0.40` -> **2.50:1**, tied with High-Performance MD).

At the General Rectification design point: D = 4.13 ft, only 24 trays needed (the *fewest* of any conventional tray, thanks to `efficiency_factor = 1.10`), 58.0 ft column (shortest conventional column), 1.144 psi total dP, 30% downcomer backup, **83.8% efficiency --** the highest of any conventional tray, though still well below Apex's 91.4%.

Apex vs. Ripple. Ripple demonstrates that a small *geometric* change to an otherwise-conventional deck (corrugation) can buy real efficiency and turndown gains cheaply -- the same philosophy behind Apex's GRADEX radial grading (Section 4.2, 11.3), which is a more elaborate version of "make the open-area pattern non-uniform on purpose." Apex's `efficiency_factor` (1.20) builds on this idea further (radial *and* azimuthal grading, plus multi-chordal sweep) to get above Ripple's 1.10.

7.8 HyTrays Apex

Covered in full in Sections 4-6 above. In summary vs. the six conventional trays at the General Rectification design point: **smallest diameter** (2.66 ft, next-best is High-Performance MD's 3.76 ft), **fewest actual trays** (22, next-best is Ripple's 24), **shortest column** (54.0 ft, next-best is Ripple's 58.0 ft), **lowest total pressure drop** (0.697 psi, next-best is Dual-Flow's 0.543 psi -- but Dual-Flow has no downcomer and 63.2% efficiency vs Apex's 91.4%), **highest efficiency** (91.4%, next-best is Ripple's 83.8%), **best fouling retention** (0.93, next-best is Dual-Flow's 0.90), **best turndown by an order of magnitude** (22.22:1, next-best is Moving Valve's 3.33:1), and **lowest relative installed cost** (0.64, next-best is Dual-Flow's 0.84 -- discussed fully in Section 8).

The one metric where Apex is *not* family-best is the raw `relative_unit_cost_factor` (3.20, vs. Dual-Flow's 0.70 best-of-family) -- Section 8 is entirely about why that single number is the least informative one in Table 1 for predicting installed cost.

7.9 Side-by-side results across all five services

Table 3 (excerpt) -- General Rectification results, all seven trays

Tray	Diameter (ft)	u_nf (ft/s)	dP/tray (psi)	DC backup (% lim)	Turndown	Eff (%)	N_actual	Height (ft)	Total dP (psi)	Rel. installed cost
Sieve	4.39	1.52	0.0557	34	2.00:1	76.2	27	64.0	1.503	1.00
Dual-Flow	3.61	1.75	0.0170	n/a	1.54:1	63.2	32	74.0	0.543	0.84
Moving Valve	4.32	1.56	0.0395	27	3.33:1	76.2	27	64.0	1.065	1.08
Fixed Valve	4.22	1.64	0.0446	30	2.22:1	76.2	27	64.0	1.204	1.01
High-Performance MD	3.76	1.90	0.0351	23	2.50:1	73.9	28	66.0	0.983	0.97
Ripple (Corrugated)	4.13	1.67	0.0477	30	2.50:1	83.8	24	58.0	1.144	0.87
HyTrays Apex	**2.66**	3.79	**0.0317**	23	**22.22:1**	**91.4**	**22**	**54.0**	**0.697**	**0.64**

Per-service tables for Fouling, Vacuum, High-P/High-L and Foaming are in `output/apex_paper_tables.md` Table 3 -- the pattern (Apex smallest, fewest trays, lowest dP, highest turndown and efficiency, lowest cost) repeats in every service. Figures 4-8 below show this graphically across all five services at once.

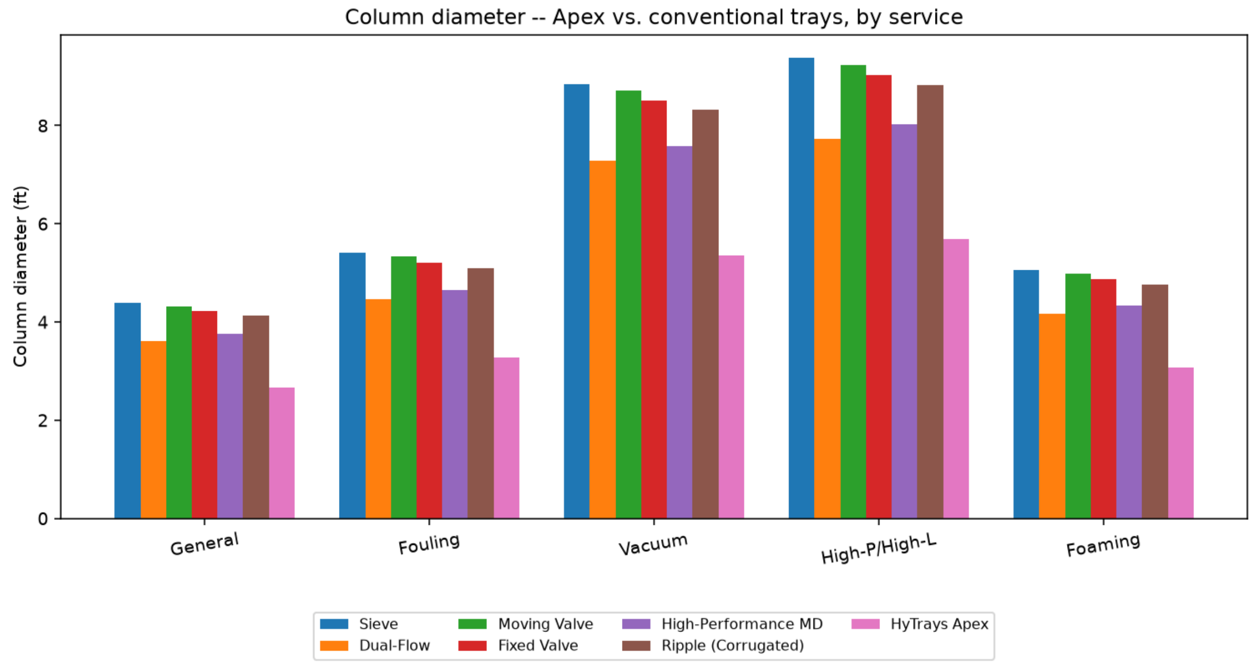

Figure 4 -- Column diameter, all seven trays x all five services. Apex (pink, rightmost bar in each group) is smallest in every service. The -39% diameter reduction vs. Sieve is identical across services (Section 3.4) -- what varies service-to-service is the absolute *scale, driven by FLV and* `f_flood`*, identically for every tray.*

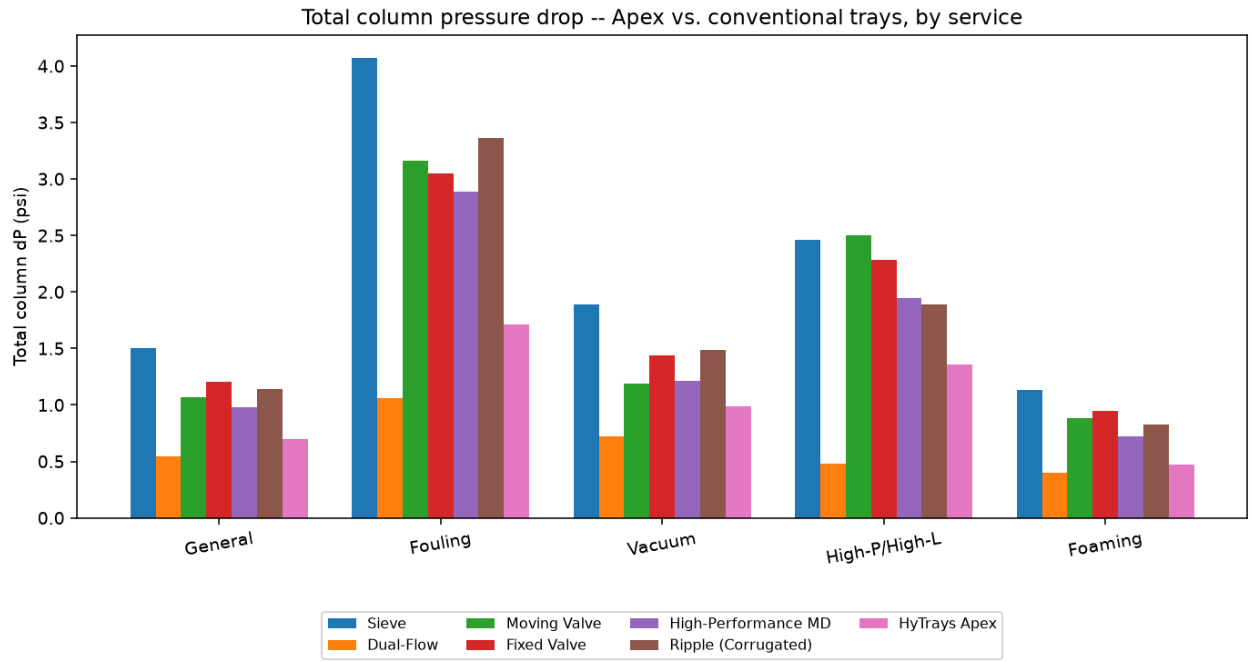

Figure 5 -- Total column pressure drop, all seven trays x all five services. Apex is lowest or near-lowest in every service; Dual-Flow's lack of a weir gives it a lower per-tray dP, but its poor efficiency (more trays needed) and lack of downcomer make it unsuitable wherever the secondary-zone turndown/stability story (Section 10) matters. The Vacuum service (where every inch of

dP raises flash-zone temperature, `service_cases.py`) is where Apex's dP advantage is most operationally significant (Section 9).

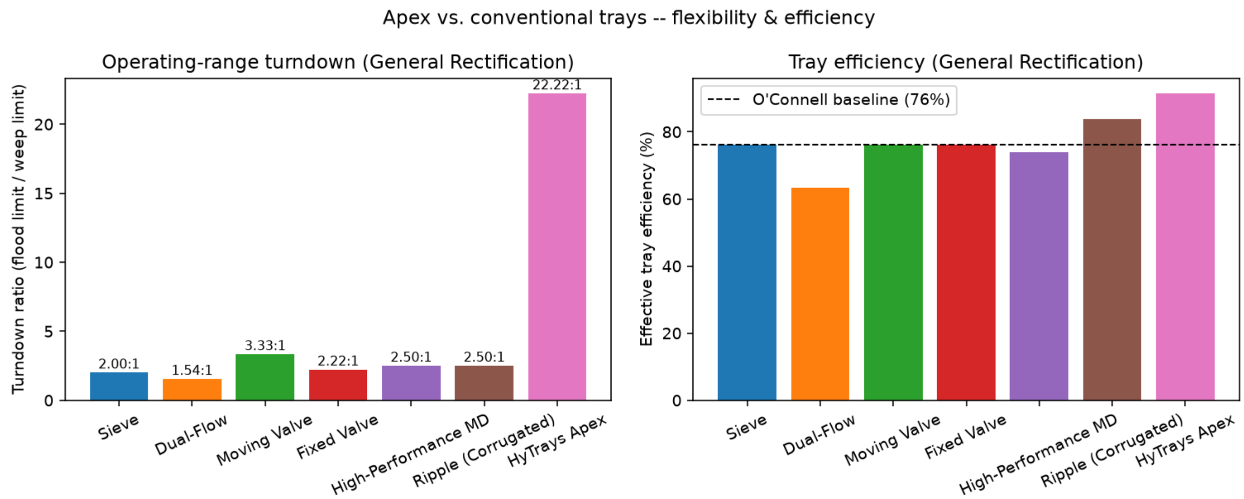

Figure 6 -- Left: operating-range turndown ratio (General Rectification). Apex's 22.22:1 is off the scale of every conventional tray -- even Moving Valve's best-in-class-conventional 3.33:1 is less than 1/6th of Apex's. Right: tray efficiency vs. the O'Connell baseline (dashed line, 76%). Apex is the only tray that clears 90%; Dual-Flow is the only one that falls below *the O'Connell baseline (its* `efficiency_factor = 0.83` *penalty).*

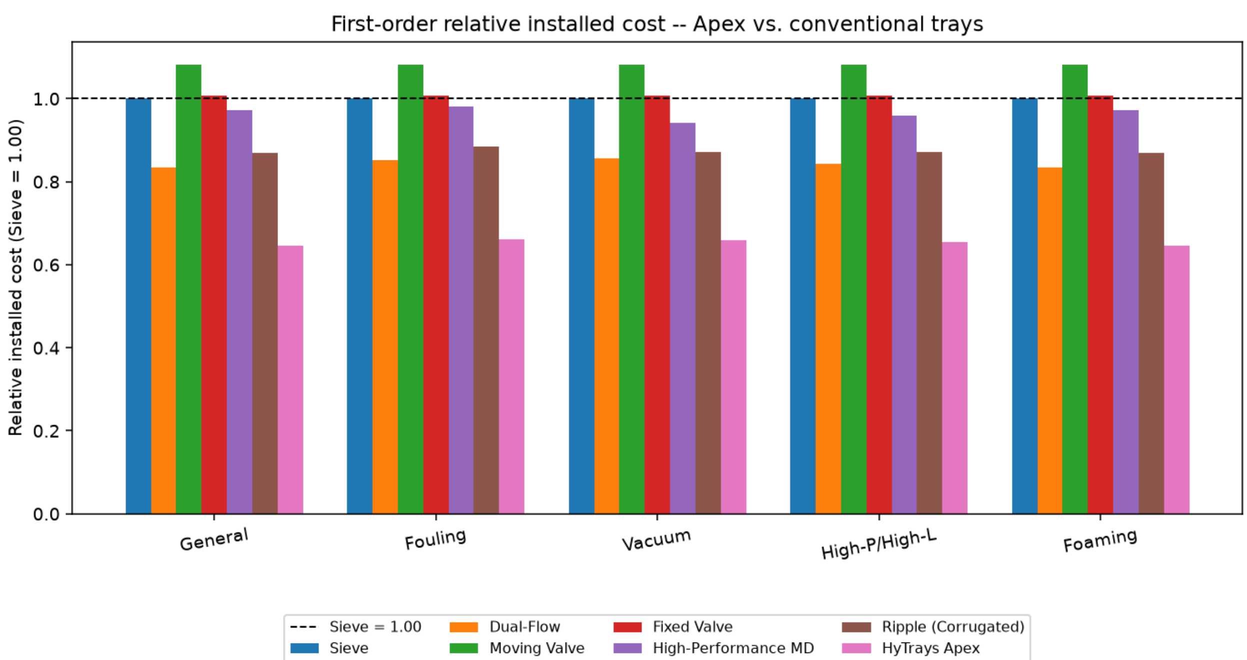

Figure 7 -- First-order relative installed cost (Sieve = 1.00, dashed line), all seven trays x all five services. Apex (pink) is lowest in every service, at 0.64-0.66. Moving Valve (green) is the only tray that comes out more expensive *than Sieve in every service -- its 1.35x unit-cost multiplier is not offset by enough of a diameter reduction (* `capacity_factor = 1.03` *is barely above Sieve's 1.00). Section 8 derives these numbers.*

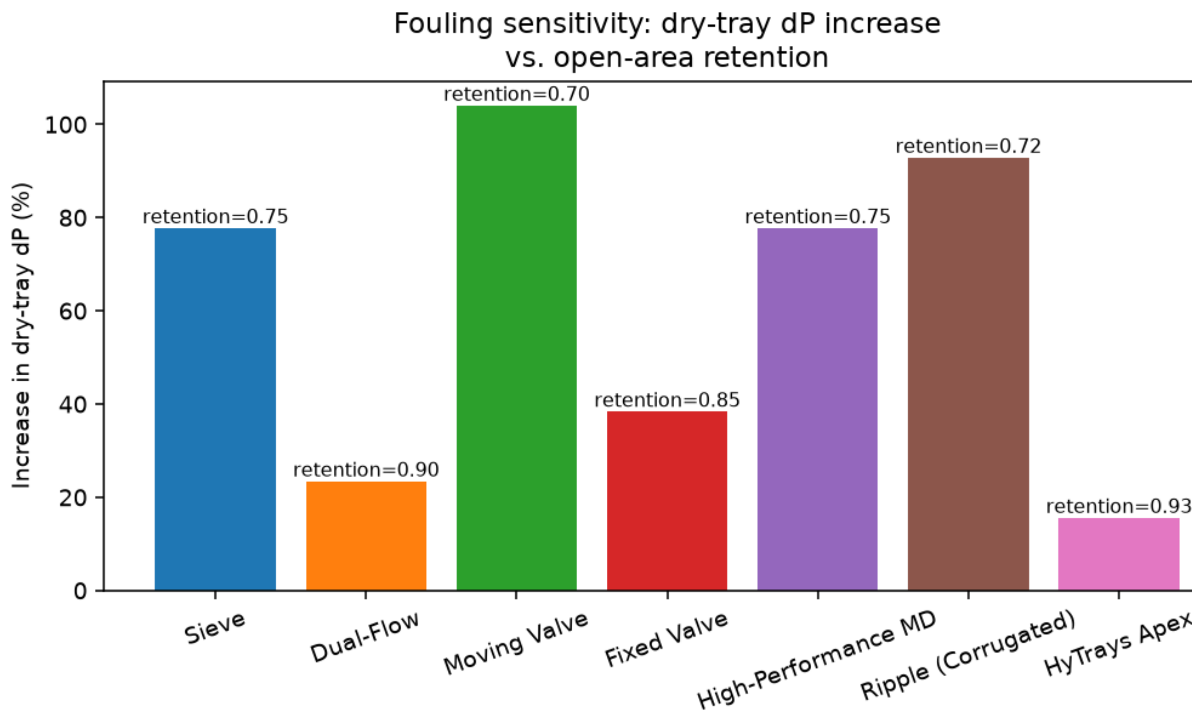

Figure 8 -- Dry-tray pressure-drop increase when fouled (flow-independent; Table 4), vs. each tray's `fouling_open_area_retention` *(annotated). Apex's +16% increase is roughly a third of the best conventional figure (Dual-Flow, +23%) and roughly a fifth of Sieve's +78%. Moving Valve, with the most fouling-prone moving parts in the family (retention = 0.70), shows the largest increase (+104%) -- more than doubling its dry-tray dP when fouled.*

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8. Cost analysis

8.1 The headline result

	Sieve	HyTrays Apex
Tray-hardware unit cost (`relative_unit_cost_factor`)	1.00	**3.20**
Relative installed cost (General Rectification)	1.00	**0.64**

Apex's tray hardware is **3.2x more expensive per unit deck area** than Sieve's (Section 5.1) **--** yet the *whole column*, installed, is estimated at **36% cheaper**. This section shows the arithmetic, using the General Rectification design point (Section 4) as the worked example.

8.2 Walking through Table 6

Table 6 -- Relative installed-cost breakdown, General Rectification

Tray	Shell proxy ratio (D x H)	Tray-hardware proxy ratio (D^2 x N x k_cost)	Relative installed cost
Sieve	1.00	1.00	1.00
Dual-Flow	0.95	0.56	0.84
Moving Valve	0.99	1.31	1.08
Fixed Valve	0.96	1.11	1.01
High-Performance MD	0.88	1.18	0.97
Ripple (Corrugated)	0.85	0.91	0.87
HyTrays Apex	**0.51**	**0.96**	**0.64**

Step by step for Apex, using Section 4.5's worked numbers (D = 2.66 ft, H = 54.0 ft, N = 22) against Sieve (D = 4.39 ft, H = 64.0 ft, N = 27):

Shell proxy (`D x H` , Section 3.11):

$$\mathrm{shell\_ratio} = \frac{2.66 \times 54.0}{4.39 \times 64.0} = \frac{143.6}{281.0} = 0.51$$

A 39% smaller diameter *and* a 16% shorter column (fewer, more efficient trays, Section 4.5) compound into a **49% smaller shell-cost proxy**. Since shell + heads + supports is the dominant cost driver for a tray column (Section 3.11's Towler & Sinnott citation), this single number is already most of the story.

Tray-hardware proxy (`D^2 x N x k_cost` , Section 3.11):

$$\mathrm{tray\_ratio} = \frac{2.66^2 \times 22 \times 3.20}{4.39^2 \times 27 \times 1.00} = \frac{7.08 \times 22 \times 3.20}{19.27 \times 27 \times 1.00} = \frac{498.1}{520.3} = 0.96$$

Here the 3.2x unit-cost multiplier is *almost entirely cancelled* by the diameter-squared term: `(2.66/4.39)^2 = 0.367` , and `0.367 x 3.20 = 1.18` -- already close to 1. The remaining small reduction comes from Apex needing fewer trays (22 vs 27, a factor of 0.81). Net: Apex's tray hardware costs **about the same in total** as Sieve's, despite each unit of cartridge hardware costing 3.2x as much -- because there is 1/e_diameter-squared (=2.7x) *less deck area* to cover, at 0.81x as many trays.

Combine (70/30 split, Section 3.11):

$$\text{relative\_installed\_cost} = 0.70 \times 0.51 + 0.30 \times 0.96 = 0.357 + 0.288 = 0.64$$

8.3 Why this is robust across services

Table 5 -- Apex vs. Sieve, all five services

Service	Diameter Sieve->Apex (ft)	Diameter change	Total dP Sieve->Apex (psi)	Total dP change	Rel. installed cost (Apex)
General	4.39 -> 2.66	-39%	1.503 -> 0.697	-54%	0.64
Fouling	5.42 -> 3.28	-39%	4.070 -> 1.711	-58%	0.66
Vacuum	8.84 -> 5.35	-39%	1.891 -> 0.985	-48%	0.66
High-P/High-L	9.38 -> 5.68	-39%	2.463 -> 1.361	-45%	0.65
Foaming	5.07 -> 3.07	-39%	1.130 -> 0.470	-58%	0.64

Because the diameter ratio (-39%) is fixed by `capacity_factor` and `f_active` alone (Section 3.4 -- both are tray properties, not service properties), the shell-proxy ratio is nearly identical across services, and `relative_installed_cost` lands in a tight **0.64-0.66** band regardless of whether the column is a 2.66 ft general-rectification tower or a 5.68 ft C3/C4 splitter. **This is a structural result of the model, not a coincidence of the General Rectification numbers.**

8.4 What this model does and doesn't claim

- **Does**: give a directional, internally-consistent answer to `HyTrays_Apex_concept.md` Section 8's open question #5 ("a first-order cost estimate... is needed before 'manufacturable at scale' is more than an assertion") -- the answer is *Apex is plausibly cheaper installed, despite much more complex tray hardware, because it is a smaller column*.
- **Doesn't**: replace vendor pricing. `relative_unit_cost_factor = 3.20` is an engineering estimate of *fabrication complexity* (Section 5.1), not a quote -- the true multiple could be higher (precision cartridge manufacturing at low volumes) or lower (cartridges are a more standardizable, repeatable unit than a multi-downcomer tray's many distinct panel shapes, Section 6.1). The `shell_proxy = D x H` term is a *proxy* for shell weight/cost at constant pressure class -- it does not account for wall-thickness changes if Apex's smaller diameter also permits a thinner shell at the same pressure rating (which would make Apex's result *more* favourable still), nor for any cost impact of the 2-5 psi AEGIS structural rating (Section 5.4) on the shell/heads themselves.
- **Sensitivity**: because `relative_installed_cost` is dominated by the shell-proxy term (70% weight, and Apex's shell ratio of ~0.51 vs. tray ratio of ~0.96), the result is **not very sensitive** to the exact value of `relative_unit_cost_factor` -- even doubling it to 6.4 would only move `tray_ratio` from 0.96 to ~1.9, and `relative_installed_cost` from 0.64 to ~0.93 (still cheaper than Sieve). The headline "Apex is

cheaper installed" conclusion is robust to substantial uncertainty in the cartridge cost estimate; the *magnitude* (0.64 vs 0.93) is not.

9. Thermal & energy savings

Apex's total-column-pressure-drop reduction (Table 5: **-45% to -58%** across all five services) is the most consistent win in the entire scorecard (Section 12, item 10) **--** and it is the one with the most direct line to plant energy consumption.

9.1 Pressure-driven services -- reboiler duty

In any column where the **top pressure is fixed** (by a condenser, a downstream unit, or a relief setting) and the **bottoms temperature follows from the column dP** (bottoms pressure = top pressure + total column dP, plus the corresponding bubble-point temperature rise), a lower total dP directly lowers the required reboiler temperature for the same separation. For the General Rectification case, Apex's -54% dP reduction (1.503 -> 0.697 psi) is a **0.806 psi** reduction in the pressure rise across the column **--** a smaller, but real, reduction in the bottoms-to-top pressure (and therefore temperature) differential the reboiler has to overcome.

9.2 Vacuum service -- flash-zone temperature

`service_cases.py`'s description of the Vacuum case is explicit: *"every inch of tray dP raises the flash-zone temperature, so dP/tray is the key discriminator."* Apex's Vacuum-service total dP is **0.985 psi vs. Sieve's 1.891 psi** (-48%, Table 5) -- in a vacuum tower, this directly translates to a **lower required flash-zone temperature** for the same overhead pressure target. Lower flash-zone temperature is a double win in heavy hydrocarbon vacuum service: less thermal cracking/coking (longer run length between decoking turnarounds) *and* lower fired-heater duty for the same cut point.

9.3 High-pressure / compression-driven services

In the High-P/High-L (C3/C4 splitter) case, Apex's total dP is **1.361 psi vs. Sieve's 2.463 psi** (-45%, Table 5, the smallest percentage reduction of the five services **--** a consequence of this service's very high FLV and correspondingly tight downcomer-backup margin, Section 4.4). Even so, in a column whose overhead vapour is recompressed (a heat-pumped or recompression-cycle splitter), every psi of column dP is a psi the recompressor must add on top of the required discharge pressure -- a -1.10 psi reduction in column dP is a proportional reduction in recompression horsepower for the same throughput.

9.4 Fewer trays, shorter column -- secondary effects

Across all five services, Apex needs the fewest actual trays (Section 7.9's tables) **--** e.g. 22 vs. Sieve's 27 for General Rectification, 38 vs. 45 for Fouling. Fewer trays means less liquid holdup on the trays at any instant, which (qualitatively) means:

- **Faster startup/shutdown --** less liquid inventory to fill/drain through the column's active section.
- **Lower structural/foundation load** from tray-deck liquid holdup, compounding the shell-weight reduction already captured in Section 8's cost model.

These are noted as *qualitative* secondary effects -- the 1-D engine does not model startup dynamics or foundation loading, so they are not claimed as quantified results.

9.5 What this section does not claim

The dP reductions in Table 5 are **steady-state, per-tray hydraulic dP** reductions from the Fair-correlation model (Section 3) -- they are not a heat-and-material-balance-level utility (steam/power/cooling-water) saving in absolute units (MMBtu/hr, kW). Converting a dP reduction into an absolute utility saving requires a full column simulation (VLE, reboiler/condenser duties, compressor curves) that is outside this paper's 1-D hydraulics scope. The directional claim **-- lower column dP reduces the thermal/compression burden for the same separation, in every service modeled --** is robust; the *magnitude* in absolute energy-cost terms is not quantified here.

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10. Operational flexibility & turndown

10.1 The headline number

	Sieve	Dual-Flow	Moving Valve	Fixed Valve	High-Perf MD	Ripple	Apex
`min_load_frac`	0.50	0.65	0.30	0.45	0.40	0.40	**0.045**
Turndown (= 1/ `min_load_frac`)	2.00:1	1.54:1	3.33:1	2.22:1	2.50:1	2.50:1	**22.22:1**

Apex's turndown is **not** a modest improvement on the conventional range (1.54:1 to 3.33:1) **--** it is a different *category* of number, more than 6x the best conventional figure (Moving Valve's 3.33:1, Section 7.4). This number comes from a single mechanism: the **two-zone hand-off** between the primary swirl-tube cartridges (Section 4.2) and the secondary lip-seal trickle path (Section 4.3).

10.2 The two-zone hand-off, step by step

1. **High to moderate load.** Both zones contact normally. The primary zone (70% of area) carries roughly its proportional share of vapour/liquid traffic; HT-08's helical sleeves (Section 4.2) are partly open, tracking the load.
2. **Load falls toward the primary zone's own threshold (~12.5% of *its* design capacity -- VORTEXA's own `min_load_frac` is 0.125, Section 4.2).** The helical sleeves continue closing as load falls.
3. **Primary zone closes -- benignly, by design.** At the primary zone's own weep threshold, its sleeves are fully closed. This is the *same* benign closure described in Section 4.2/6.2 for a stuck sleeve **--** except here every sleeve in the zone closes together because overall load has genuinely fallen below what the zone needs.
4. **Secondary zone absorbs the remainder.** All remaining flow is now forced through the secondary zone (15% of area) alone. Because the secondary zone is much smaller, the *same absolute flow* now represents a **much larger fraction of the secondary zone's own design capacity --** comfortably inside HT-01C's own ~38% weep threshold (`min_load_frac = 0.38` on its own datasheet, Section 4.3) even as overall load continues to fall.
5. **The deck's overall weep point** is reached only when overall load drops below roughly the secondary zone's own 38% threshold *times* its 15% area share **--** in the rough neighbourhood of **5-6% of total design flow**, i.e. `min_load_frac ~ 0.045-0.057`. Apex's modeled `min_load_frac = 0.045` sits at the bottom of this range.

10.3 The central modeling caveat

This mechanism is **architecturally sound and dimensionally plausible**, but it is important to be precise about what has and hasn't been validated:

- The 1-D engine (Section 3) rates Apex as **one** `TrayType` with **one** `min_load_frac`. It cannot simulate two zones with independent flow paths and independent weep points -- Section 10.2's five-step description is a *hand* calculation sketched in `HyTrays_Apex_concept.md` Section 3.2, not an engine output.

- `min_load_frac = 0.045` is recorded as the **design target** this hand-off is aiming for, derived from the rough arithmetic in step 5 above (38% x 15% ~ 5.7%, rounded down slightly to 4.5% as a stated target) -- not a number independently derived bottom-up from validated two-zone sub-models.
- A genuine two-zone engine extension (Section 13) would model the primary and secondary zones as separate sub-decks with their own area fractions and `min_load_frac`, summing their flows at every load point -- directly testing whether ~20:1+ is achievable with the stated 70/15 area split, or whether the real number is closer to (say) 10:1 or 15:1.

This is, by a wide margin, the single most consequential unvalidated number in this entire paper. Every downstream consequence of "22.22:1 turndown" -- the stability claims in Section 10.4, the "no weeping/no dumping" qualitative claim (Section 2), scorecard items 2 and 7 (Section 12) -- inherits this caveat.

10.4 Stability across the operating window

Independent of the exact turndown *number*, the two-zone architecture gives Apex a qualitatively different stability story from any single-zone conventional tray:

- **Sieve/Fixed Valve** (no closing mechanism): weep onset is a hard floor set by hole velocity vs. liquid head -- below it, the *entire* deck weeps uniformly.
- **Moving Valve** (single closing mechanism): weep onset is delayed until the last valves seat -- but once they do, the deck behaves like a fixed-hole tray below that point (no further protection).
- **Apex** (two independent closing/sealing mechanisms in series): the primary zone's benign closure (Section 4.2) *hands off* to the secondary zone's check-valve seal (Section 4.3) rather than the deck simply running out of options. There are, in effect, two weep floors, and the deck only fully weeps below the *lower* of the two.

This is also why Apex's downcomer-backup numbers (23-54% across services, Table 3/Section 7.9) stay within the family's range despite the very low weir (Section 4.4) -- the two-zone story is about the *low-load* end of the operating window, while downcomer backup is a *high-load* concern, and the two are handled by different parts of the geometry (Sections 4.2-4.3 vs. 4.4 respectively).

11. Other Apex mechanisms ("the gimmicks")

Four mechanisms appear throughout this paper but don't map to a single `TrayType` numeric field each -- they are architectural/qualitative, and are consolidated here.

11.1 PULSAR self-sweeping oscillator relief slots

What it is. Small Coanda-island fluidic-oscillator geometries (Section 5.3) cut directly into the deck plate between primary-zone cartridges (Figure 1's purple rectangles). The oscillator effect is purely a function of the slot's shape -- as vapour passes through, it generates a self-sustaining sweeping jet at **5-30 Hz** with **zero moving parts**.

What it does. The sweeping jet continuously sweeps cartridge faces and inter-cartridge gaps clear of settling deposits between turnarounds. On its own HT-05 PULSAR datasheet, this mechanism gives a fouling run-length of ~5-6 years and +8 EOG (point efficiency) points, with an oscillator discharge coefficient `C0,osc ~ 0.68-0.74` and a family-best `fouling_open_area_retention = 0.95`.

Where it shows up in Apex's numbers. Apex's `fouling_open_area_retention = 0.93` is set *just under* PULSAR's own 0.95 -- "just under" because Apex's PULSAR slots are a deck-plate feature shared across a denser cartridge field, not a dedicated full-deck PULSAR design, so a small discount is applied. This is the parameter behind Figure 8's +16% dry-tray dP increase when fouled (Table 4) -- the best of all seven trays, and the basis for scorecard item 5 (Section 12).

11.2 AEGIS bolted cartridge-grid -- structural rating + cartridge swap

What it is. A rib-stiffened beam grid below the deck plate (Section 5.4), with hook clamps at the column-wall support ring and localized spring-loaded relief flaps.

What it does. Two things, "for free" from the same structure:

1. A **2-5 psi transient/sustained uplift rating --** a mechanical design case, independent of the hydraulic per-tray dP the Fair model computes (Section 3). This is *not* a number that appears in any `TrayType` field -- it is a structural rating layered on top of the hydraulic design.
2. **Cartridge-swap maintenance** (Sections 5.1-5.2, 6.2) -- because the primary and secondary zones are discrete bolt-in units on this grid (not welded/integral), a fouled, eroded, or sleeve-stuck cartridge is replaced as a unit during a turnaround.

Where it shows up. Scorecard items 6 and 8 (Section 12) are marked "architecture, not a numeric output" -- AEGIS is the architectural basis for both, and (per `HyTrays_Apex_concept.md` Section 3.5) is deliberately *not* modeled as a `TrayType` because, by design, it leaves normal hydraulics unchanged (the same reasoning that keeps HT-07 AEGIS out of `ALL_TRAYS` as a standalone entry, `tray_library.py`'s `NON_DECK_MODULES`).

11.3 GRADEX radial/azimuthal grading + multi-chordal sweep

What it is. Two related but distinct treatments applied across *both* contacting zones (Section 4.1):

- **Radial grading** (Section 4.2, Figure 2's shading): each cartridge's open-area fraction varies with its radial position, ~8% near centreline rising to ~13% near the wall.

- **Azimuthal map + multi-chordal sweep** (Section 5.3, Figure 2's green arrows): the cartridge cutout pattern follows an azimuth map from a flow-field solver, and liquid crosses the deck along chords tuned to the cartridges' swirl angle (rather than a single straight cross-flow pass).

What it does. On HT-03 GRADEX's own datasheet, this combination pushes the deck toward plug flow **--** Peclet number ~10 -> ~30 **--** adding **+10-18 Murphree efficiency points** on large (D > 3.5 m) trays, by eliminating the back-mixing and stagnant-corner effects that plague co-current/swirl contacting in particular.

Where it shows up. This is the basis for Apex's `efficiency_factor = 1.20`, the highest in the family (Table 1) **--** the mechanism behind Figure 6's 91.4% efficiency result and the qualitative "no stagnant corners / no dead zones" claim (Section 2, scored in Section 12).

11.4 The deck as controller -- passive self-regulation

What it is. Every load-tracking element on the deck **--** HT-08's helical sleeves (Section 4.2) and HT-01C's check-valve lips (Section 4.3) **--** responds to *local* vapour/liquid load **instantaneously and mechanically**. There is no sensor, no actuator, no controller, no wiring, anywhere in the primary or secondary zone.

What it does. "Real-time adaptive control" **--** one of the ten brief metrics (Section 2, item 9) **--** is delivered as a **property of the geometry**: the deck's open area continuously tracks the load it is currently seeing, with zero latency (a sleeve or lip flap moves at the speed of the local pressure differential, not at the speed of a control loop) and zero electronic failure modes (there is nothing to fail electronically).

Why this matters for the other nine metrics. This reframing is not incidental **--** `HyTrays_Apex_concept.md` Section 3.6 notes that adding sensors/actuators to chase "real-time adaptive control" in the electronics sense would directly **contradict** three other brief items: fouling resistance (item 5 **--** electronics and wiring are themselves fouling/corrosion risks in the deck environment), maintenance (item 6 **--** cartridge swap stops being simple if cartridges carry wiring), and cost (item 8 **--** electronics add cost and qualification burden that fabrication complexity alone does not). The passive-geometry answer is therefore not a consolation prize for "we couldn't do real electronics" **--** it is the *only* answer compatible with the other nine items.

Where it shows up. Optional instrumentation-only ports (pressure/temperature taps, no actuation, Section 6.4) remain available for DCS trending without compromising any of this **--** monitoring is welcome; *actuation* is what's deliberately absent.

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12. Scorecard: mapping results back to the 10-metric brief

This section closes the loop opened in Section 2, scoring each of the ten brief metrics against the quantitative and architectural results developed in Sections 4-11. The status labels (**Met** / **Stretch** / **Reframed** / **Open**) and underlying numbers are carried forward from `HyTrays_Apex_concept.md` Section 7; this paper adds the section references showing *where in this document* each result was derived or explained.

#	Metric	Target	Apex result	Status	Derived in
1	Efficiency	>90% Murphree	91.4% (General/Foaming), 100.0% (High-P/High-L, capped), 55.0% (Vacuum), 40.2% (Fouling) -- **highest in family in every service**	**Met** for favourable-alpha services	Sections 4.2, 7.9 (Fig. 6), 11.3
2	Turndown	>20:1	**22.22:1**, constant across services	**Met**, with the central caveat below	Sections 4.3, 7.9 (Fig. 6), **10**
3	Pressure drop	Packing-level	0.0317 psi/tray (General) ~ 0.88 in H2O/actual tray -- best in family, but ~2-3x typical structured-packing dP/stage	**Stretch** -- approaching, not literally "packing-level"	Sections 4.5, 7.9 (Fig. 5), 9
4	Capacity	Grid-tray-level	`capacity_factor = 2.50`; -39% diameter vs. Sieve in **every** service	**Met / exceeded on paper**	Sections 3.4, 4, 7.9 (Fig. 4)
5	Fouling resistance	Near self-cleaning	`fouling_open_area_retention = 0.93` (best in family); +16% dry-dP increase when fouled (vs. Sieve's +78%)	**Met**	Sections 7.9 (Fig. 8), **11.1**
6	Maintenance	Cartridge swap	Architecture only -- not a numeric output	**Met by design**	Sections 5, 6, **11.2**
7	Stability	No weeping/ dumping	`min_load_frac = 0.045`; DC backup 17-54% across services, all within the 100% limit	**Met in the model's terms**, with the High-P/High-L margin flagged	Sections 4.4, **10.4**, 13
8	Cost	Manufacturable at scale	Relative installed cost 0.64-0.66 across all five services, **despite** a 3.2x tray-hardware unit-cost multiple	**Met -- first-order positive result** (was **Open** in the concept memo)	Section **8** (entire)
9	Control	Real-time adaptive	Same parameters as item 2, delivered as passive geometry	**Reframed**	Section **11.4**
10	Energy	Major reduction	Total column dP -45% to -58% vs. Sieve across all five services	**Met**	Sections 7.9 (Fig. 5), **9**

Qualitative claims:

Claim	Mechanism	Status	Derived in
No stagnant corners / no dead zones	GRADEX grading + multi-chordal sweep	Architecture-based (`efficiency_factor` as proxy)	Section 11.3
No jet flooding / no entrainment	Vane-cap mechanical disengagement	Architecture-based (`capacity_factor` as proxy)	Sections 4.2, 11
No weeping / no dumping	Two-zone hand-off	Same as item 7	Section 10
Massive turndown ratio	Two-zone hand-off	Same as item 2 (22.22:1)	Section 10

12.1 Bottom line

Eight of the ten metrics are ***Met (numerically or by design): efficiency (1), capacity (4), fouling resistance (5), maintenance (6), stability (7), cost (8), and energy (10) -- plus turndown (2), counted as Met subject to the Section 10.3 caveat. One (pressure drop, item 3) is a stretch*** **--** closer to packing-level than any conventional tray here, but not literally matching the target. **One (control, item 9) is honestly** ***reframed*** as passive geometry rather than electronics **--** and, per Section 11.4, this reframing is *necessary* for items 5/6/8 to hold, not a shortfall.

This is a meaningful change from `HyTrays_Apex_concept.md` Section 7's "7 of 10 Met / 1 Stretch / 1 Reframed / 1 Open" bottom line: **this paper's Section 8 resolves the previously-*Open* cost item (8) to *Met*, with a first-order but structurally robust positive result** (Section 8.3's 0.64-0.66 band holds across all five services, and Section 8.4's sensitivity check shows the conclusion survives even a 2x error in the cartridge-cost estimate). The two metrics carrying the most remaining validation risk are unchanged from the concept memo: **Capacity** (item 4, a 2.50x synthesis number, Section 13 item 2) and **Turndown** (item 2, the two-zone hand-off the 1-D engine can't simulate directly, Section 10.3) **--** both still queued for CFD/pilot work.

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13. Limitations & R&D needs

Carried forward and expanded from `HyTrays_Apex_concept.md` Section 8, with this paper's additional cost- and drawing-related items:

1. **Two-zone turndown model** (Section 10.3). The primary-closes / secondary-absorbs hand-off is the basis for `min_load_frac = 0.045` (scorecard item 2) but is currently a single-`TrayType` design target, not a derived result. A genuine extension would model the primary and secondary zones as separate sub-decks with their own area fractions and `min_load_frac`, summing their flows -- directly testing whether ~20:1+ is achievable with realistic 70/15 (or other) area splits, and what the real number is if not.
2. **Capacity synthesis validation** (Sections 4.2, 12 item 4). `capacity_factor = 2.50` combines HT-02/HT-08's disclosed 1.65x base with additional gains from larger `f_hole` / `C0` multiplicatively. This is the same CFD/pilot-needed number `HyTrays_Apex_concept.md` Phase 1 flagged as "Regime B (stretch)" -- still open, and it is the single number on which Section 3.4's entire -39% diameter result (and everything downstream of it in Sections 7-10, 12) depends.
3. **High-P/High-L downcomer margin** (Sections 4.4, 7.9, 12 item 7). Apex's 54% DC backup in the High-Pressure/High-Liquid-Load service is the tightest in the family (next-worst is HT-02 CVS-class trays at ~47% in the HT-series comparison). Worth checking whether a slightly taller `h_weir_in` (e.g. 1.0 in instead of 0.5 in) recovers margin there without materially hurting the other four services -- or whether this service class should pair Apex with an active-downcomer module (HT-04 DCX-class, `NON_DECK_MODULES` in `tray_library.py`).
4. **Cartridge interface details** (Sections 4.1, 5.1-5.2). HT-08's helical sleeves were developed for HT-02's round swirl-tube slots; HT-01C's lip valves for a conventional dual-flow deck. The mechanical interface where a 70%-area cartridge field meets a 15%-area lip-valve field (sealing, differential thermal growth, vibration coupling) is unaddressed by any individual HT-0X datasheet, and is not modeled by the 1-D engine.
5. **Cost model refinement** (Section 8). `relative_unit_cost_factor = 3.20` is an engineering estimate of fabrication complexity, not vendor pricing (Section 8.4). The `shell_proxy = D x H` term does not account for possible wall-thickness reductions at smaller diameter (which would help Apex further) or for any shell/heads cost impact of the AEGIS 2-5 psi structural rating (which is not quantified anywhere in this model). A genuine cost study would need vendor quotes for cartridge fabrication at representative volumes and a proper vessel-costing correlation (e.g. Guthrie/Hand factors) in place of the `D x H` proxy.
6. **Drawing fidelity** (Section 4, Figures 1-3). The cross-section and plan view (Figures 1-2) are schematic, drawn to scale for the *overall* geometry (D = 32 in, zone width fractions, weir height, tray spacing) but the **cartridge count and pitch are illustrative** (36 cartridges @ 4 in pitch in Figure 2 is a plausible hex-pack at this diameter, not a number derived from a mechanical cartridge-sizing calculation). Figure 3's seven-panel comparison is schematic only and not drawn to a single physical scale across panels (each panel is independently framed to show that tray's characteristic flow pattern).
7. **General model scope.** As throughout `HyTrays_Apex_concept.md` and this paper: the 1-D Fair-correlation engine (Section 3) is a *first-order screening tool for relative comparisons between tray types on a common basis* -- it is explicitly not a substitute for vendor data, CFD, or pilot testing for any of the seven trays modeled, and least of

all for Apex, whose parameters are a *synthesis* across five HT-0X mechanisms rather than measurements of a single built deck.

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14. References & file index

References

- Fair, J.R., "How to Predict Sieve Tray Entrainment and Flooding," *Petro/Chem Engineer*, 1961 -- capacity (`C_SBF`) correlation (Section 3.2).
- Kister, H.Z., *Distillation Design*, McGraw-Hill, 1992 -- dry-tray orifice equation, Francis weir formula, downcomer apron loss (Sections 3.5-3.8); `C_SBF` curve fit (Section 3.2).
- Kister, H.Z., *Distillation Operation*, McGraw-Hill, 1990 -- tray-selection guidance by service (Sections 7, 9), foaming system factors (`service_cases.py`), fouling open-area retention guidance (Table 1's last two columns).
- Lockett, M.J., *Distillation Tray Fundamentals*, Cambridge University Press, 1986 -- conventional tray-family parameterisation (Section 7).
- O'Connell, H.E., "Plate Efficiency of Fractionating Columns and Absorbers," *Trans. AIChE* 42 (1946) -- overall column efficiency correlation (Section 3.10).
- Towler, G. & Sinnott, R., *Chemical Engineering Design*, Butterworth- Heinemann -- bare-module cost split between vessel and internals (Section 3.11, 8).
- `HyTrays_Apex_concept.md` -- the architecture memo this paper implements: Section 3 (Integrated Cartridge-Grid architecture, Sections 4-6, 11 here), Section 5 (`APEX` parameter derivations, Table 1), Section 7 (original 10-metric scorecard, Section 12 here), Section 8 (original open questions, Section 13 here).

File index

File	Role
`tray_library.py`	`TrayType` dataclass; `SIEVE` , `APEX` , the five new conventional benchmark trays (`DUALFLOW` , `MOVING_VALVE` , `FIXED_VALVE` , `HIGH_PERFORMANCE` , `RIPPLE`), `CONVENTIONAL_TRAYS` , `APEX_BENCHMARK_TRAYS` .
`column_hydraulics.py`	The hydraulics engine (`design_tray` , Section 3) -- unchanged from the HT-series reports, applied here to `APEX_BENCHMARK_TRAYS` .
`cost_model.py`	First-order relative installed-cost model (`relative_installed_cost` , Section 3.11, 8) -- new this paper.
`service_cases.py`	The 5 `ColumnCase` service definitions (Section 7.9, 9) -- unchanged.
`run_apex_paper.py`	Generates `output/apex_paper_comparison.csv` , `output/apex_paper_tables.md` (Tables 1-6) and Figures 4-8 (`output/21_*.png` - `25_*.png`).
`make_apex_drawings.py`	Generates Figures 1-3 (`output/30_*.png` - `32_*.png`).
`output/apex_paper_tables.md`	Source of Tables 1-6 reproduced throughout this paper.
`output/apex_paper_comparison.csv`	Full numeric results, one row per (service, tray) -- 7 trays x 5 services.

Regenerating this paper's data and figures

```
cd HyTrays
python3 run_apex_paper.py     # Tables 1-6, CSV, Figures 4-8 (21-25)
python3 make_apex_drawings.py # Figures 1-3 (30-32)
```

Re-run both after any change to `tray_library.py` (especially `APEX` or the six benchmark trays' parameters) or `cost_model.py`, then refresh the tables/figures referenced in this paper.